Sr No.	Topic	Date	Sign
	Module 1:		
1.	a. Installing Red Hat Enterprise Linux (Virtual Box/ VmWare)		
	b. Linux Graphical Environment i. Display/Login Manager ii. Desktop Environment iii. Linux Directory Structure and File Systems		
2.	Basic System Commands		
	a. Use input-output redirection (>, >>, |, 2>, etc)		
	b. Use grep and regular expressions to analyse text		
	c. Log in and switch users in multi-user targets		
	d. Working with Text Files cat,cut,less,hesd,tail,sort,wc,grep, awk		
3.	Essential File Management Tools		
	a. Managing Files ,Using Links		
	b. Working with Archives and Compressed Files		
	c. Create, delete, copy, and move files and directories		
	d. Archive, compress, unpack, and uncompress files using tar, star, gzip, and bzip2		
	e. Create hard and soft links		
	f. Managing File Ownership, Managing Basic Permissions		
	g. Managing Advanced Permissions, Managing ACLs		
	h. Setting Default Permissions with umask		
	i. Working with User-Extended Attributes		
	j. List, set, and change standard ugo/rwx permissions		
	k. Create and configure set-GID directories for collaboration		

	l. Create and manage access control lists		
4.	Working with Users, Groups, and Permissions		
	a. Creating and Managing Users		
	b. Creating and Managing Groups		
	c. Using Permissions and Advanced Permissions		
	d. Change passwords and adjust password aging for local user accounts		
	e. Configure superuser access -> su,sudo,root		
5.	a. Managing Software Packages with YUM		
	b. Using yum		
	c. Managing Software Packages with RPM		
	d. Install and update software packages repository from the local file system		
	Module 2:		
6.	Managing Processes		
	a.Introduction to Process Management		
	b. Managing Shell Jobs		
	c. Using Common Command-Line Tools for Process Management		
	d. Using top to Manage Processes		
	e. Using tuned to Optimize Performance		
	f. Identify CPU/memory-intensive processes and kill processes		
7.	Managing Storage		
	a. Understanding MBR and GPT Partitions		
	b. Managing Partitions and File Systems		
	c. Mounting File Systems		
	d. List, create, delete partitions on MBR and GPT disks		
	e. Configure systems to mount file systems at boot by universally		

	unique ID (UUID) or label		
	f. Add new partitions and logical volumes, and swap to a system non-destructively		
	g. Create, mount, unmount, and use vfat, ext4, and xfs file systems		
8.	Configuring Networking		
	a. Managing Network Addresses and Interfaces		
	b. Validating Network Configuration		
	c. Managing Network Configuration with nmtui and nmcli		
	d. Configure IPv4 and IPv6 addresses		
	e. Configure DNS-hostname resolution		
9.	Configuring SSH		
	a. Using SSH and Related Utilities		
	b. Access remote systems using SSH		
	c. Configuring Key-Based Authentication with Passphrases		
	d. Configure key-based authentication for SSH		
10.	Configuring web server		
	a. Configuring a Basic Apache Server		
	b. Understanding Apache Configuration Files		
	c. Creating Apache Virtual Host		
11.	Configure Firewall		
	a. Linux Firewall with iptables		
	b. Working with Firewalld		
12.	Accessing Network Storage		
	a. Using NFS Services		
	b. Using CIFS Services		
	c. Mounting Remote File Systems Through fstab		

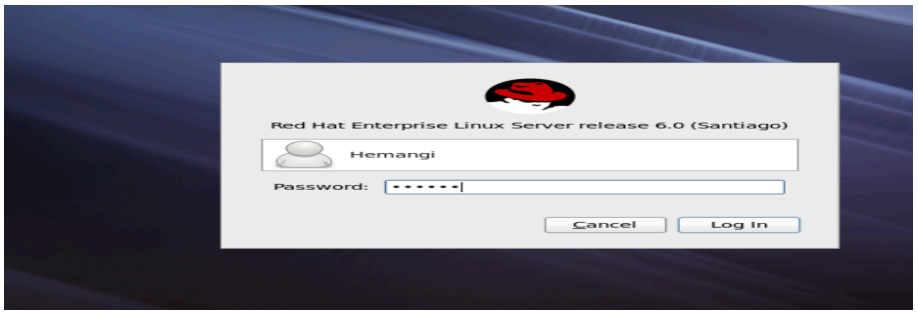
	d. Using Automount to Mount Remote File Systems		
13.	Creating Shell scripts		
	a. automate or monitor tasks/processes		
	b. Getting and setting system information		
14.	Scheduling Tasks		
	a. Configuring Cron to Automate Recurring Tasks		
	b. Configuring At to Schedule Future Tasks		
	c. Schedule tasks using at and cron		

Practical 1:

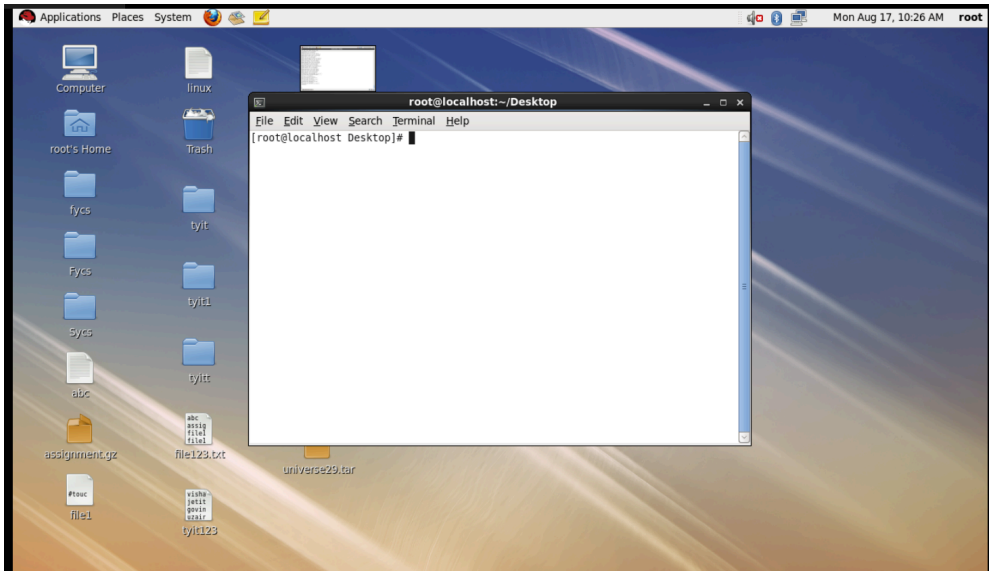
a. Installing Red Hat Enterprise Linux (Virtual Box/ VmWare).

b. Linux Graphical Environment

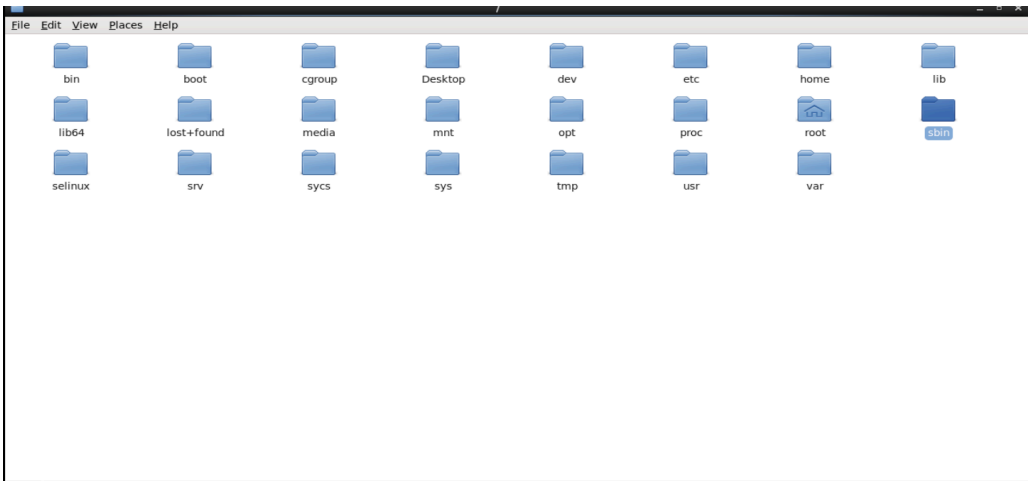
i. Display/Login Manager



ii. Desktop Environment



iii. Linux Directory Structure and File Systems



Practical 2: Basic System Commands

a. Use input-output redirection (>, >>, |, 2>, etc)

```
[root@localhost Desktop]# ls
atul  file123  file123.txt  file345  filename  fycs  fycs1  sycs  tycs  tycs1  tycs2  tyit
[root@localhost Desktop]# ls >file123.txt
[root@localhost Desktop]# ls
atul  file123  file123.txt  file345  filename  fycs  fycs1  sycs  tycs  tycs1  tycs2  tyit
[root@localhost Desktop]# cat file123.txt
atul
file123
file123.txt
file345
filename
fycs
fycs1
sycs
tycs
tycs1
tycs2
tyit
```

```
[root@localhost Desktop]# cat >tyit
bash: tyit: Is a directory
[root@localhost Desktop]# cat >tyit1234
Hemangi
^C
[root@localhost Desktop]# ls
atul  file123  file123.txt  file345  filename  fycs  fycs1  Screenshot-1.png  Screenshot.png  sycs  tycs  tycs1  tycs2  tyit  tyit123  tyit1234
[root@localhost Desktop]#  cat >>tyit1234
123456
^C
[root@localhost Desktop]# cat tyit1234
Hemangi
123456
[root@localhost Desktop]#  cat >>tyit1234
root
^C
```

b. Use grep and regular expressions to analyse text

```
[root@localhost Desktop]# clear

[root@localhost Desktop]# rpm -qa |grep gnome
system-gnome-theme-60.0.2-1.el6.noarch
libgnomeui-2.24.1-4.el6.x86_64
gnome-backgrounds-2.28.0-2.el6.noarch
gnome-menus-2.28.0-4.el6.x86_64
NetworkManager-gnome-0.8.1-5.el6.x86_64
gnome-user-share-2.28.2-3.el6.x86_64
gnome-doc-utils-stylesheets-0.18.1-1.el6.noarch
gnome-vfs2-smb-2.24.2-6.el6.x86_64
gnome-media-2.29.91-6.el6.x86_64
gnome-python2-desktop-2.28.0-4.el6.x86_64
gnome-disk-utility-libs-2.30.1-2.el6.x86_64
gnome-applets-2.28.0-7.el6.x86_64
gnome-keyring-2.28.2-6.el6.x86_64
gnome-speech-0.4.25-3.1.el6.x86_64
gnome-python2-bonobo-2.28.0-3.el6.x86_64
gnome-desktop-2.28.2-8.el6.x86_64
gnome-panel-2.30.2-5.el6.x86_64
gnome-user-docs-2.28.0-4.el6.noarch
gnome-utils-2.28.1-10.el6.x86_64
libgnomecanvas-2.26.0-4.el6.x86_64
polkit-gnome-0.96-3.el6.x86_64
gnome-icon-theme-2.28.0-2.el6.noarch
compiz-gnome-0.8.2-24.el6.x86_64
gnome-python2-gnomekeyring-2.28.0-4.el6.x86_64
gnome-python2-extras-2.25.3-20.el6.x86_64
libgnome-2.28.0-11.el6.x86_64
gnome-python2-gnome-2.28.0-3.el6.x86_64
gnome-disk-utility-ui-libs-2.30.1-2.el6.x86_64
gnome-python2-applet-2.28.0-4.el6.x86_64
gnome-session-2.28.0-15.el6.x86_64
gnome-session-xsession-2.28.0-15.el6.x86_64
gnome-disk-utility-2.30.1-2.el6.x86_64
gnome-packagekit-2.28.3-3.el6.x86_64
gnome-system-monitor-2.28.0-7.el6.x86_64
```

c. Log in and switch users in multi-user targets

```
Red Hat Enterprise Linux ...

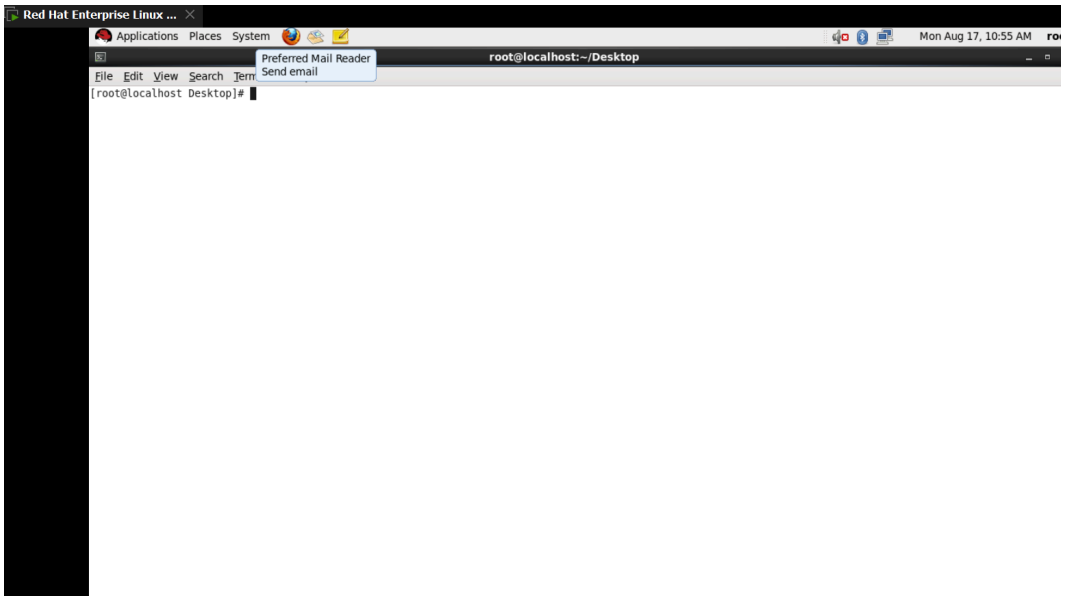
Red Hat Enterprise Linux Server release 6.0 (Santiago)
Kernel 2.6.32-71.el6.x86_64 on an x86_64

localhost login: root
Password:
[root@localhost ~]# _
```

```
Red Hat Enterprise Linux ...

Red Hat Enterprise Linux Server release 6.0 (Santiago)
Kernel 2.6.32-71.el6.x86_64 on an x86_64

localhost login: admin
Password:
[admin@localhost ~]$ _
```



d. Working with Text Files cat,cut,less,hesd,tail,sort,wc,grep, awk

```
[root@localhost Desktop]# cat >tyit1234
^C
[root@localhost Desktop]# head tyit1234
[root@localhost Desktop]# tail tyit1234
[root@localhost Desktop]#
```

Practical 3: Essential File Management Tools

a. Managing Files ,Using Links
b. Working with Archives and Compressed Files
c. Create, delete, copy, and move files and directories
d. Archive, compress, unpack, and uncompress files using tar, star, gzip, and bzip2
e. Create hard and soft links
f. Managing File Ownership, Managing Basic Permissions
g. Managing Advanced Permissions, Managing ACLs
h. Setting Default Permissions with umask
i. Working with User-Extended Attributes
j. List, set, and change standard ugo/rwx permissions
k. Create and configure set-GID directories for collaboration
l. Create and manage access control lists

```
[root@localhost linux_lab]# mkdir linux_lab
[root@localhost linux_lab]# cd linux_lab
[root@localhost linux_lab]# touch file1.txt file2.txt file3.txt
[root@localhost linux_lab]# mkdir docs backup
[root@localhost linux_lab]# cp file1.txt docs/
[root@localhost linux_lab]# mv file2.txt docs/
[root@localhost linux_lab]# mv file3.txt notes.txt
[root@localhost linux_lab]# ln notes.txt hardlink.txt
[root@localhost linux_lab]# ln -s notes.txt softlink.txt
[root@localhost linux_lab]# ls -li
total 8
667396 drwxr-xr-x. 2 root root 4096 Sep  9 11:04 backup
667395 drwxr-xr-x. 2 root root 4096 Sep  9 11:05 docs
667392 -rw-r--r--. 1 root root    0 Sep  9 11:04 file1.txt
667394 -rw-r--r--. 2 root root    0 Sep  9 11:04 hardlink.txt
667394 -rw-r--r--. 2 root root    0 Sep  9 11:04 notes.txt
667398 lrwxrwxrwx. 1 root root    9 Sep  9 11:06 softlink.txt -> notes.txt
[root@localhost linux_lab]# tar -cvf backup.tar docs/
docs/
docs/file2.txt
docs/file1.txt
[root@localhost linux_lab]# tar -czvf backup.tar.gz docs/
docs/
docs/file2.txt
docs/file1.txt
[root@localhost linux_lab]# mkdir extracted
[root@localhost linux_lab]# tar -xzvf backup.tar.gz-C extracted/
tar (child): backup.tar.gz-C: Cannot open: No such file or directory
tar (child): Error is not recoverable: exiting now
tar: Child returned status 2
tar: Error is not recoverable: exiting now
[root@localhost linux_lab]# chmod 640 notes.txt
[root@localhost linux_lab]# ls -l notes.txt
-rw-r-----. 2 root root 0 Sep  9 11:04 notes.txt
[root@localhost linux_lab]# sudo chown $USER:$USER notes.txt
[root@localhost linux_lab]# umask
0022
```

```
0022
[root@localhost linux_lab]# umask 027
[root@localhost linux_lab]# mkdir collaboration
[root@localhost linux_lab]# chmod 2775 collaboration
[root@localhost linux_lab]# ls -ld collaboration
drwxrwsr-x. 2 root root 4096 Sep  9 11:13 collaboration
[root@localhost linux_lab]# setfacl -m u:$USER:rwx collaboration
[root@localhost linux_lab]# getfacl collaboration
# file: collaboration
# owner: root
# group: root
# flags: -s-
user::rwx
user:root:rwx
group::rwx
mask::rwx
other::r-x

[root@localhost linux_lab]# lsattr notes.txt
-------------e- notes.txt
[root@localhost linux_lab]#
```

Practical 4: Working with Users, Groups, and Permissions

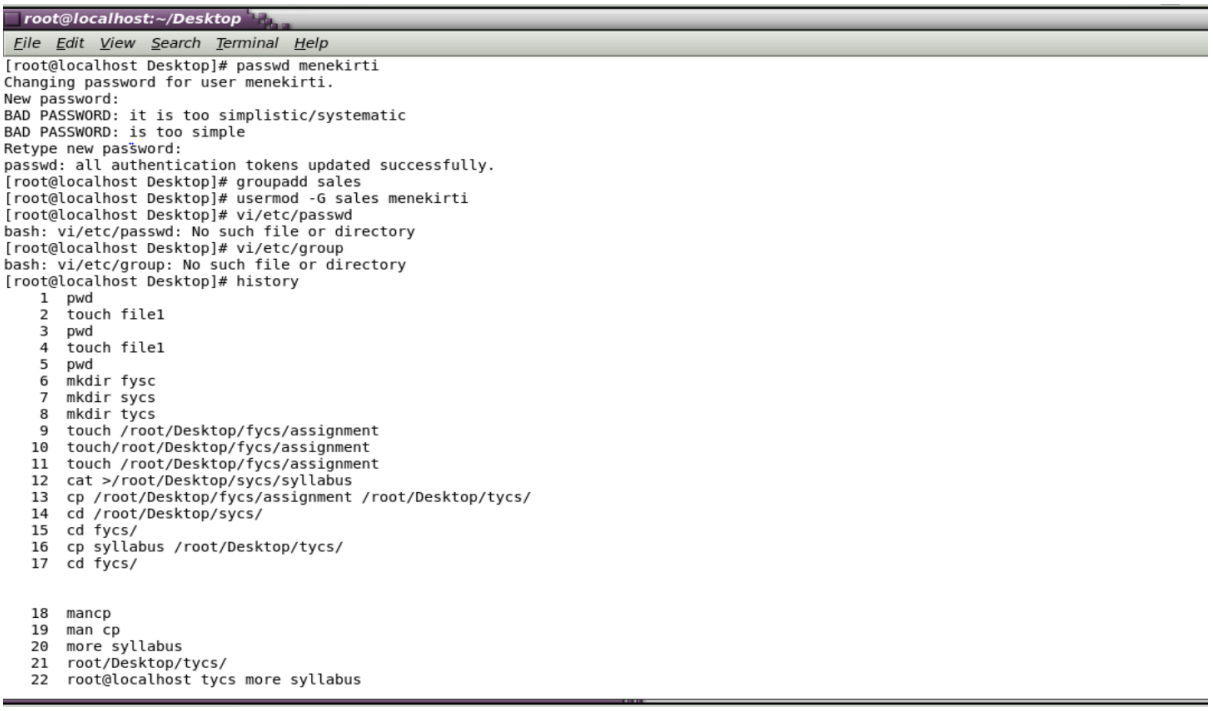
a. Creating and Managing Users

b. Creating and Managing Groups

c. Using Permissions and Advanced Permissions

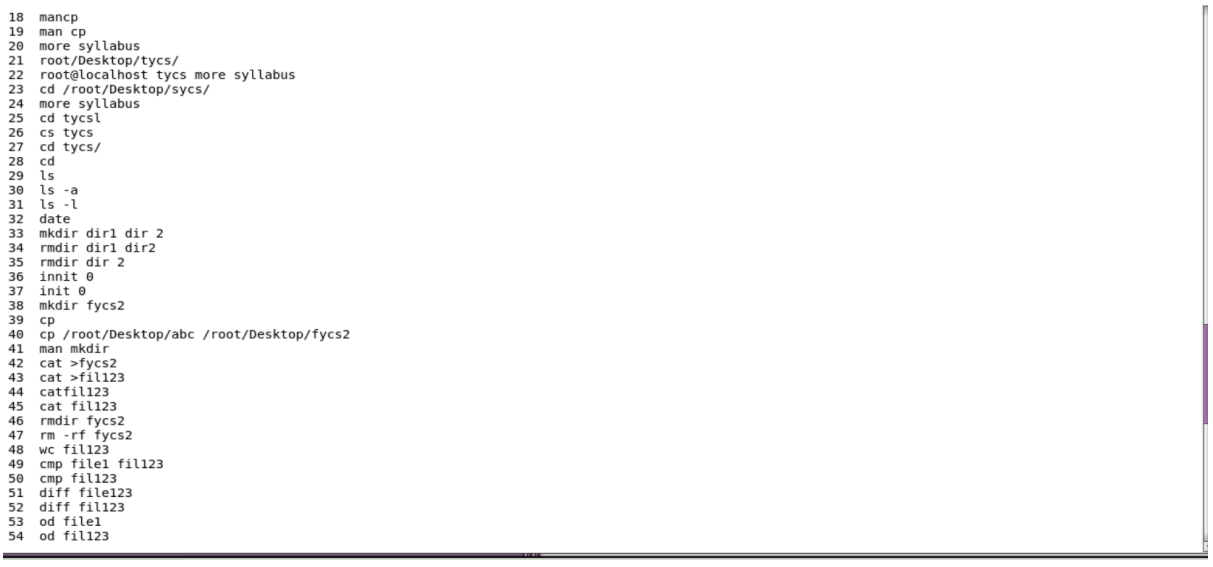
d. Change passwords and adjust password aging for local user accounts

e. Configure superuser access -> su,sudo,root



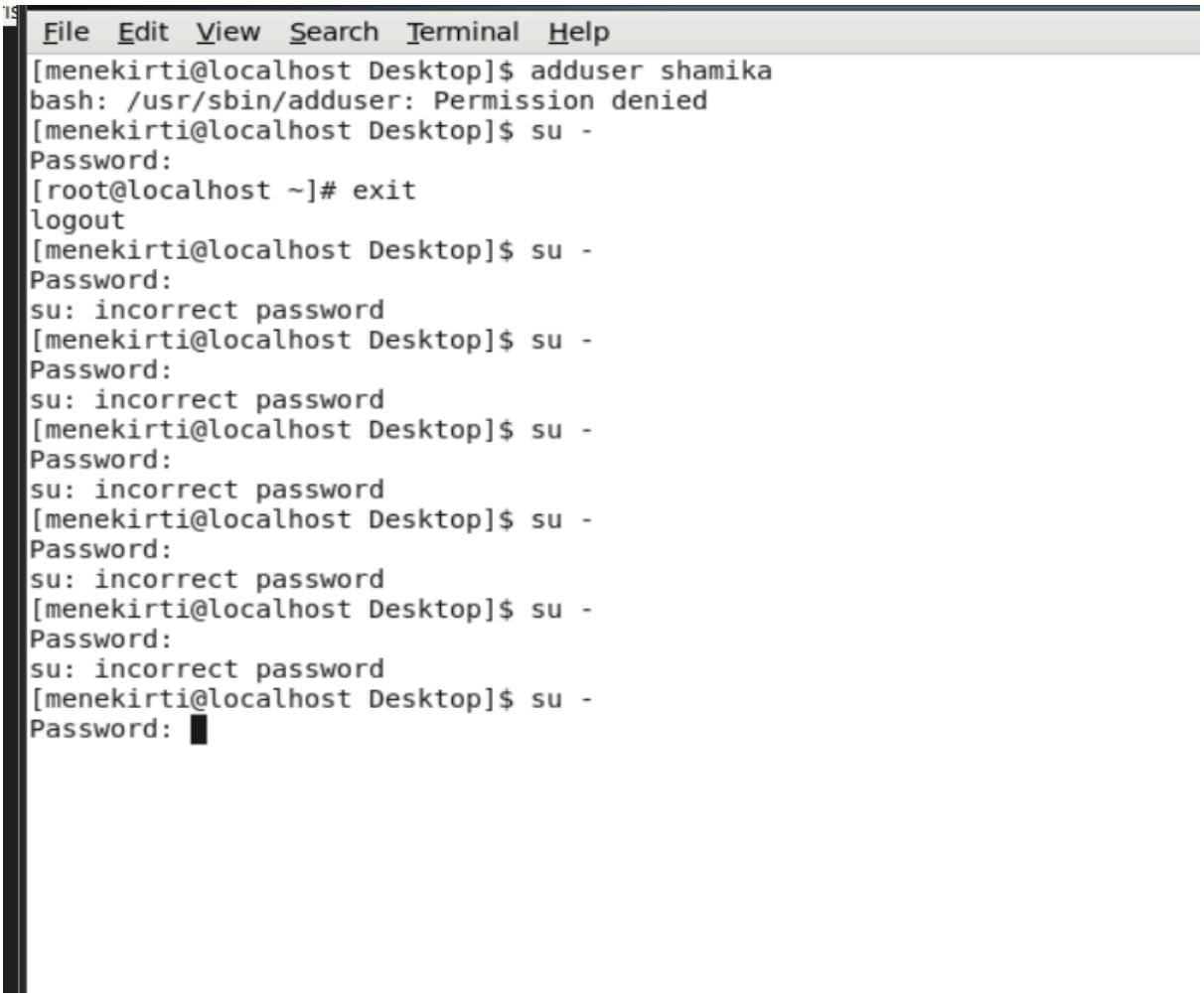
```
root@localhost:~/Desktop
File  Edit  View  Search  Terminal  Help
[root@localhost Desktop]# passwd menekirti
Changing password for user menekirti.
New password:
BAD PASSWORD: it is too simplistic/systematic
BAD PASSWORD: is too simple
Retype new password:
passwd: all authentication tokens updated successfully.
[root@localhost Desktop]# groupadd sales
[root@localhost Desktop]# usermod -G sales menekirti
[root@localhost Desktop]# vi/etc/passwd
bash: vi/etc/passwd: No such file or directory
[root@localhost Desktop]# vi/etc/group
bash: vi/etc/group: No such file or directory
[root@localhost Desktop]# history
    1  pwd
    2  touch file1
    3  pwd
    4  touch file1
    5  pwd
    6  mkdir fysc
    7  mkdir sycs
    8  mkdir tycs
    9  touch /root/Desktop/fycs/assignment
   10  touch/root/Desktop/fycs/assignment
   11  touch /root/Desktop/fycs/assignment
   12  cat >/root/Desktop/sycs/syllabus
   13  cp /root/Desktop/fycs/assignment /root/Desktop/tycs/
   14  cd /root/Desktop/sycs/
   15  cd fycs/
   16  cp syllabus /root/Desktop/tycs/
   17  cd fycs/


   18  mancp
   19  man cp
   20  more syllabus
   21  root/Desktop/tycs/
   22  root@localhost tycs more syllabus
```



```
18  mancp
19  man cp
20  more syllabus
21  root/Desktop/tycs/
22  root@localhost tycs more syllabus
23  cd /root/Desktop/sycs/
24  more syllabus
25  cd tycsl
26  cs tycs
27  cd tycs/
28  cd
29  ls
30  ls -a
31  ls -l
32  date
33  mkdir dir1 dir 2
34  rmdir dir1 dir2
35  rmdir dir 2
36  innit 0
37  init 0
38  mkdir fycs2
39  cp
40  cp /root/Desktop/abc /root/Desktop/fycs2
41  man mkdir
42  cat >fycs2
43  cat >fil123
44  catfil123
45  cat fil123
46  rmdir fycs2
47  rm -rf fycs2
48  wc fil123
49  cmp file1 fil123
50  cmp fil123
51  diff file123
52  diff fil123
53  od file1
54  od fil123
```

```
drwxr-xr-x.  22 root root  4096 Jul 28  2025 var
[root@localhost Desktop]# chmod 666 /mydir
[root@localhost Desktop]# ls -l /
total 98
dr-xr-xr-x.   2 root root  4096 Aug 11  2025 bin
dr-xr-xr-x.   5 root root  1024 Jul 28  2025 boot
drwxr-xr-x.   2 root root  4096 Jul 14  2010 cgroup
drwxr-xr-x.  17 root root  3740 Aug  7 04:08 dev
drwxr-xr-x.  98 root root  4096 Aug  7 04:24 etc
drwxr-xr-x.   7 root root  4096 Aug  7 04:22 home
dr-xr-xr-x.  10 root root  4096 Jul 28  2025 lib
dr-xr-xr-x.   9 root root 12288 Aug 11  2025 lib64
drwx------.   2 root root 16384 Jul 28  2025 lost+found
drwxr-xr-x.   2 root root  4096 Dec  4  2009 media
drwxr-xr-x.   3 root root  4096 Jul 28  2025 mnt
drw-rw-rw-.   2 root root  4096 Aug  7 04:34 mydir
drwxr-xr-x.   2 root root  4096 Jul 28  2025 opt
dr-xr-xr-x. 177 root root     0 Aug  7 04:08 proc
dr-xr-x---.  29 root root  4096 Aug  7 04:10 root
dr-xr-xr-x.   2 root root 12288 Aug 11  2025 sbin
drwxr-xr-x.   7 root root     0 Aug  7 04:08 selinux
drwxr-xr-x.   2 root root  4096 Dec  4  2009 srv
drwxr-xr-x.  13 root root     0 Aug  7 04:08 sys
drwxrwxrwt.  31 root root  4096 Aug  7 04:18 tmp
drwxr-xr-x.  13 root root  4096 Jul 28  2025 usr
drwxr-xr-x.  22 root root  4096 Jul 28  2025 var
[root@localhost Desktop]# chmod 664 /mydir
[root@localhost Desktop]# ls -l /
total 98
dr-xr-xr-x.   2 root root  4096 Aug 11  2025 bin
dr-xr-xr-x.   5 root root  1024 Jul 28  2025 boot
drwxr-xr-x.   2 root root  4096 Jul 14  2010 cgroup
drwxr-xr-x.  17 root root  3740 Aug  7 04:08 dev
[illegible]
root@localhost Desktop]# chmod o-w /mydir/
root@localhost Desktop]# ls -ls /
otal 98
4 dr-xr-xr-x.   2 root root  4096 Aug 11  2025 bin
2 dr-xr-xr-x.   5 root root  1024 Jul 28  2025 boot
4 drwxr-xr-x.   2 root root  4096 Jul 14  2010 cgroup
0 drwxr-xr-x.  17 root root  3740 Aug  7 04:08 dev
4 drwxr-xr-x.  98 root root  4096 Aug  7 04:24 etc
4 drwxr-xr-x.   7 root root  4096 Aug  7 04:22 home
4 dr-xr-xr-x.  10 root root  4096 Jul 28  2025 lib
2 dr-xr-xr-x.   9 root root 12288 Aug 11  2025 lib64
6 drwx------.   2 root root 16384 Jul 28  2025 lost+found
4 drwxr-xr-x.   2 root root  4096 Dec  4  2009 media
4 drwxr-xr-x.   3 root root  4096 Jul 28  2025 mnt
4 drw-rw-r--.   2 root root  4096 Aug  7 04:34 mydir
4 drwxr-xr-x.   2 root root  4096 Jul 28  2025 opt
0 dr-xr-xr-x. 177 root root     0 Aug  7 04:08 proc
4 dr-xr-x---.  29 root root  4096 Aug  7 04:10 root
2 dr-xr-xr-x.   2 root root 12288 Aug 11  2025 sbin
0 drwxr-xr-x.   7 root root     0 Aug  7 04:08 selinux
4 drwxr-xr-x.   2 root root  4096 Dec  4  2009 srv
0 drwxr-xr-x.  13 root root     0 Aug  7 04:08 sys
4 drwxrwxrwt.  31 root root  4096 Aug  7 04:45 tmp
4 drwxr-xr-x.  13 root root  4096 Jul 28  2025 usr
4 drwxr-xr-x.  22 root root  4096 Jul 28  2025 var
root@localhost Desktop]#
```



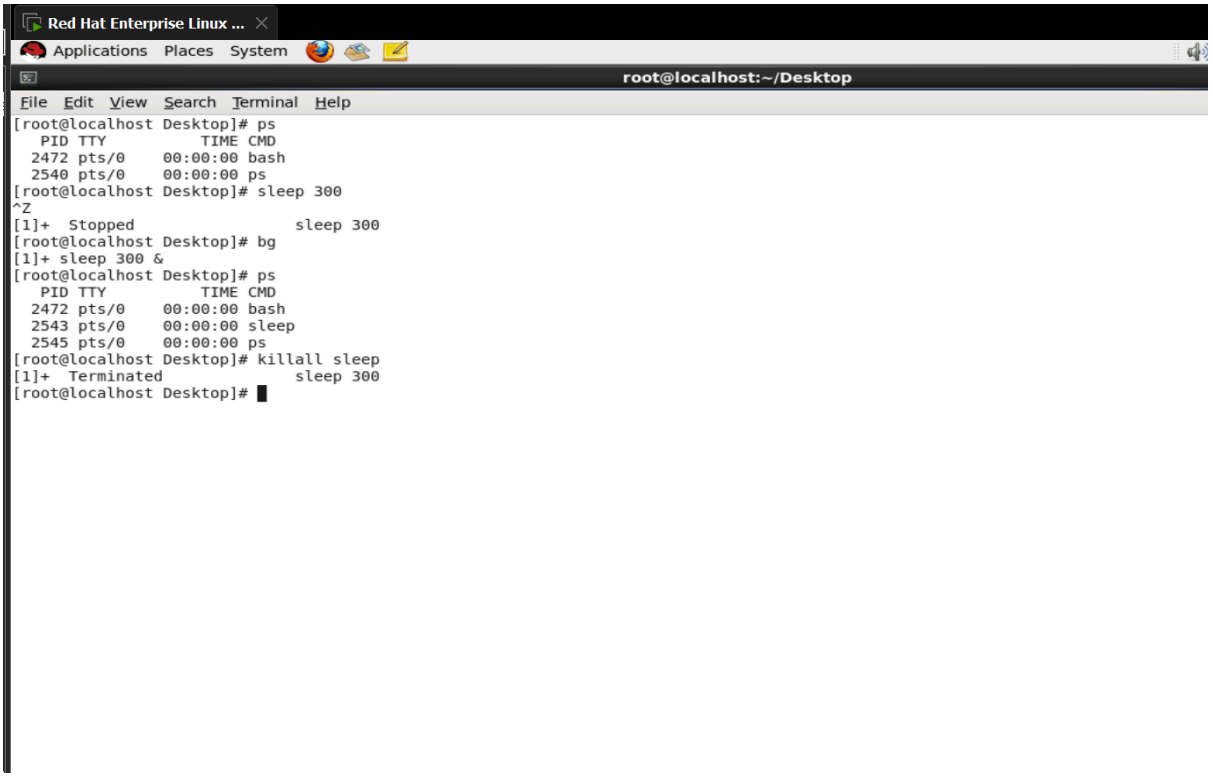

```
drwxrwxrwt.  31 root root  4096 Aug  7 04:18 tmp
drwxr-xr-x.  13 root root  4096 Jul 28  2025 usr
drwxr-xr-x.  22 root root  4096 Jul 28  2025 var
[root@localhost Desktop]# chmod 664 /mydir
[root@localhost Desktop]# ls -l /
total 98
dr-xr-xr-x.   2 root root  4096 Aug 11  2025 bin
dr-xr-xr-x.   5 root root  1024 Jul 28  2025 boot
drwxr-xr-x.   2 root root  4096 Jul 14  2010 cgroup
drwxr-xr-x.  17 root root  3740 Aug  7 04:08 dev
drwxr-xr-x.  98 root root  4096 Aug  7 04:24 etc
drwxr-xr-x.   7 root root  4096 Aug  7 04:22 home
dr-xr-xr-x.  10 root root  4096 Jul 28  2025 lib
dr-xr-xr-x.   9 root root 12288 Aug 11  2025 lib64
drwx------.   2 root root 16384 Jul 28  2025 lost+found
drwxr-xr-x.   2 root root  4096 Dec  4  2009 media
drwxr-xr-x.   3 root root  4096 Jul 28  2025 mnt
drw-rw-r--.   2 root root  4096 Aug  7 04:34 mydir
drwxr-xr-x.   2 root root  4096 Jul 28  2025 opt
dr-xr-xr-x. 177 root root     0 Aug  7 04:08 proc
dr-xr-x---.  29 root root  4096 Aug  7 04:10 root
dr-xr-xr-x.   2 root root 12288 Aug 11  2025 sbin
drwxr-xr-x.   7 root root     0 Aug  7 04:08 selinux
drwxr-xr-x.   2 root root  4096 Dec  4  2009 srv
drwxr-xr-x.  13 root root     0 Aug  7 04:08 sys
drwxrwxrwt.  31 root root  4096 Aug  7 04:18 tmp
drwxr-xr-x.  13 root root  4096 Jul 28  2025 usr
drwxr-xr-x.  22 root root  4096 Jul 28  2025 var
[root@localhost Desktop]# chmod ou- /mydir
[root@localhost Desktop]# ls -l /
total 98
dr-xr-xr-x.   2 root root  4096 Aug 11  2025 bin
dr-xr-xr-x.   5 root root  1024 Jul 28  2025 boot
drwxr-xr-x.   2 root root  4096 Jul 14  2010 cgroup
drwxr-xr-x.  17 root root  3740 Aug  7 04:08 dev
drwxr-xr-x.  98 root root  4096 Aug  7 04:24 etc
drwxr-xr-x.   7 root root  4096 Aug  7 04:22 home
dr-xr-xr-x.  10 root root  4096 Jul 28  2025 lib
```

Practical5
Aim: Managing Software
a. Managing Software Packages with YUM
b. Using yum
c. Managing Software Packages with RPM
d. Install and update software packages repository from the local file
System

Module2:
Practical 6
Aim: introduction to process management
A process is a running instance of a program. Every process has a unique PID (process id)

Useful concepts:

Term	meaning
PID	unique ID assigned to a process
PPID	Parent process ID
UID	User who owns the process
CPU%	percentage of CPU being used
MEM%	percentage of memory being used
Proce**ss state**	**Running, sleep, stopped, zombie etc**

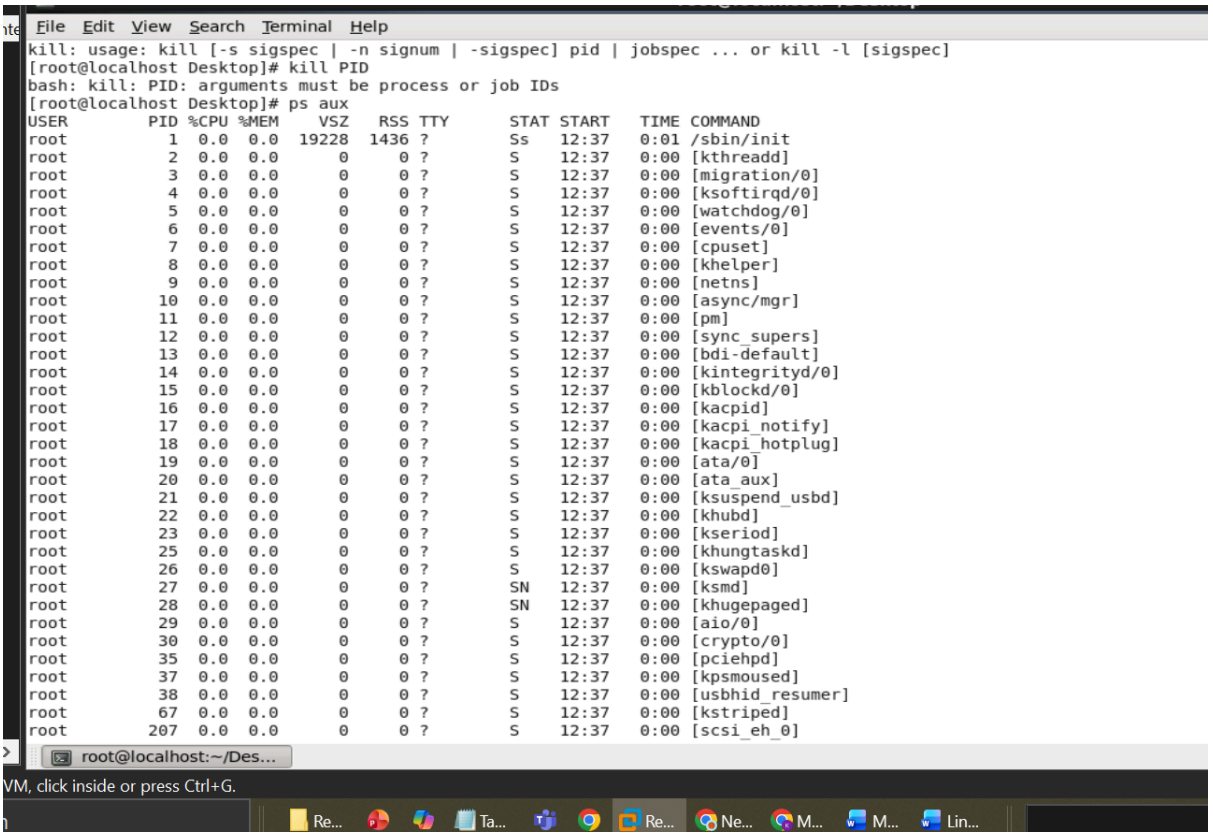


View process
Ps
Shows Processes associated with the current shell.

```
File  Edit  View  Search  Terminal  Help
[root@localhost Desktop]# ps
   PID TTY          TIME CMD
  2472 pts/0    00:00:00 bash
  2854 pts/0    00:00:00 ps
[root@localhost Desktop]#
```

Ps aux

Shows processes for all users with detailed information.



```
File  Edit  View  Search  Terminal  Help
kill: usage: kill [-s sigspec | -n signum | -sigspec] pid | jobspec ... or kill -l [sigspec]
[root@localhost Desktop]# kill PID
bash: kill: PID: arguments must be process or job IDs
[root@localhost Desktop]# ps aux
USER        PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root          1  0.0  0.0  19228  1436 ?        Ss   12:37   0:01 /sbin/init
root          2  0.0  0.0      0     0 ?        S    12:37   0:00 [kthreadd]
root          3  0.0  0.0      0     0 ?        S    12:37   0:00 [migration/0]
root          4  0.0  0.0      0     0 ?        S    12:37   0:00 [ksoftirqd/0]
root          5  0.0  0.0      0     0 ?        S    12:37   0:00 [watchdog/0]
root          6  0.0  0.0      0     0 ?        S    12:37   0:00 [events/0]
root          7  0.0  0.0      0     0 ?        S    12:37   0:00 [cpuset]
root          8  0.0  0.0      0     0 ?        S    12:37   0:00 [khelper]
root          9  0.0  0.0      0     0 ?        S    12:37   0:00 [netns]
root         10  0.0  0.0      0     0 ?        S    12:37   0:00 [async/mgr]
root         11  0.0  0.0      0     0 ?        S    12:37   0:00 [pm]
root         12  0.0  0.0      0     0 ?        S    12:37   0:00 [sync_supers]
root         13  0.0  0.0      0     0 ?        S    12:37   0:00 [bdi-default]
root         14  0.0  0.0      0     0 ?        S    12:37   0:00 [kintegrityd/0]
root         15  0.0  0.0      0     0 ?        S    12:37   0:00 [kblockd/0]
root         16  0.0  0.0      0     0 ?        S    12:37   0:00 [kacpid]
root         17  0.0  0.0      0     0 ?        S    12:37   0:00 [kacpi_notify]
root         18  0.0  0.0      0     0 ?        S    12:37   0:00 [kacpi_hotplug]
root         19  0.0  0.0      0     0 ?        S    12:37   0:00 [ata/0]
root         20  0.0  0.0      0     0 ?        S    12:37   0:00 [ata_aux]
root         21  0.0  0.0      0     0 ?        S    12:37   0:00 [ksuspend_usbd]
root         22  0.0  0.0      0     0 ?        S    12:37   0:00 [khubd]
root         23  0.0  0.0      0     0 ?        S    12:37   0:00 [kseriod]
root         25  0.0  0.0      0     0 ?        S    12:37   0:00 [khungtaskd]
root         26  0.0  0.0      0     0 ?        S    12:37   0:00 [kswapd0]
root         27  0.0  0.0      0     0 ?        SN   12:37   0:00 [ksmd]
root         28  0.0  0.0      0     0 ?        SN   12:37   0:00 [khugepaged]
root         29  0.0  0.0      0     0 ?        S    12:37   0:00 [aio/0]
root         30  0.0  0.0      0     0 ?        S    12:37   0:00 [crypto/0]
root         35  0.0  0.0      0     0 ?        S    12:37   0:00 [pciehpd]
root         37  0.0  0.0      0     0 ?        S    12:37   0:00 [kpsmoused]
root         38  0.0  0.0      0     0 ?        S    12:37   0:00 [usbhid_resumer]
root         67  0.0  0.0      0     0 ?        S    12:37   0:00 [kstriped]
root        207  0.0  0.0      0     0 ?        S    12:37   0:00 [scsi_eh_0]
root@localhost:~/Des...
VM, click inside or press Ctrl+G.
```

ps -ef
Another commonly used full

```
ot       2471  0.0  0.0   8132   672 ?        S    12:38   0:00 gnome-pty-helper
ot       2472  0.0  0.0 108248  1772 pts/0    Ss   12:38   0:00 /bin/bash -l
ot       2805  2.0  0.0 107968  1052 pts/0    R+   13:48   0:00 ps aux
u have new mail in /var/spool/mail/root
oot@localhost Desktop]#
oot@localhost Desktop]# ps -ef
D          PID   PPID  C STIME TTY          TIME CMD
ot           1      0  0 12:37 ?        00:00:01 /sbin/init
ot           2      0  0 12:37 ?        00:00:00 [kthreadd]
ot           3      2  0 12:37 ?        00:00:00 [migration/0]
ot           4      2  0 12:37 ?        00:00:00 [ksoftirqd/0]
ot           5      2  0 12:37 ?        00:00:00 [watchdog/0]
ot           6      2  0 12:37 ?        00:00:00 [events/0]
ot           7      2  0 12:37 ?        00:00:00 [cpuset]
ot           8      2  0 12:37 ?        00:00:00 [khelper]
ot           9      2  0 12:37 ?        00:00:00 [netns]
ot          10      2  0 12:37 ?        00:00:00 [async/mgr]
ot          11      2  0 12:37 ?        00:00:00 [pm]
ot          12      2  0 12:37 ?        00:00:00 [sync_supers]
ot          13      2  0 12:37 ?        00:00:00 [bdi-default]
ot          14      2  0 12:37 ?        00:00:00 [kintegrityd/0]
ot          15      2  0 12:37 ?        00:00:00 [kblockd/0]
ot          16      2  0 12:37 ?        00:00:00 [kacpid]
ot          17      2  0 12:37 ?        00:00:00 [kacpi_notify]
ot          18      2  0 12:37 ?        00:00:00 [kacpi_hotplug]
ot          19      2  0 12:37 ?        00:00:00 [ata/0]
ot          20      2  0 12:37 ?        00:00:00 [ata_aux]
ot          21      2  0 12:37 ?        00:00:00 [ksuspend_usbd]
ot          22      2  0 12:37 ?        00:00:00 [khubd]
ot          23      2  0 12:37 ?        00:00:00 [kseriod]
ot          25      2  0 12:37 ?        00:00:00 [khungtaskd]
ot          26      2  0 12:37 ?        00:00:00 [kswapd0]
ot          27      2  0 12:37 ?        00:00:00 [ksmd]
ot          28      2  0 12:37 ?        00:00:00 [khugepaged]
ot          29      2  0 12:37 ?        00:00:00 [aio/0]
ot          30      2  0 12:37 ?        00:00:00 [crypto/0]
ot          35      2  0 12:37 ?        00:00:00 [pciehpd]
ot          37      2  0 12:37 ?        00:00:00 [kpsmoused]
 root@localhost:~/Des...
```

Process states

Common states include:

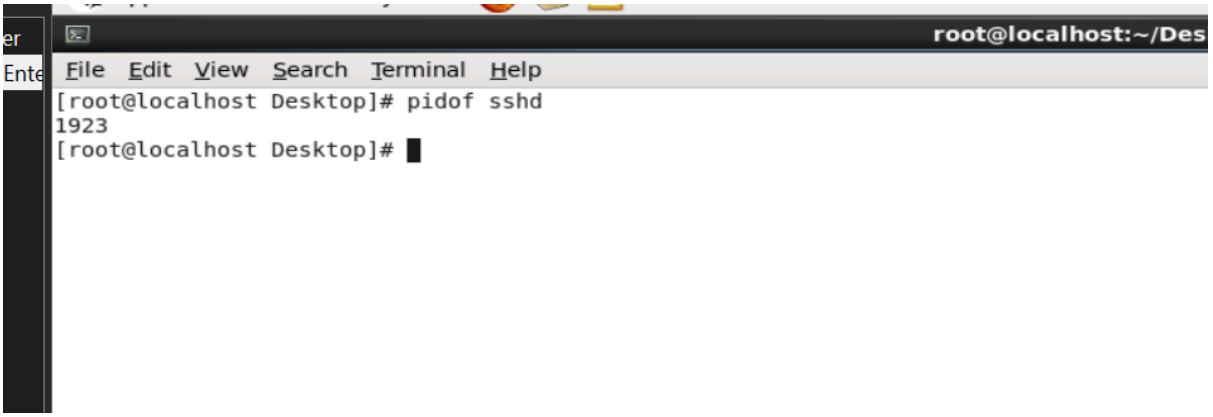
2.pgrep
Find process by name

```
[root@localhost Desktop]# pgrep
pgrep: No matching criteria specified
Usage: pgrep [-flvx] [-d DELIM] [-n|-o] [-P PPIDLIST] [-g PGRPLIST] [-s
        [-u EUIDLIST] [-U UIDLIST] [-G GIDLIST] [-t TERMLIST] [PATTERN]
[root@localhost Desktop]#
```

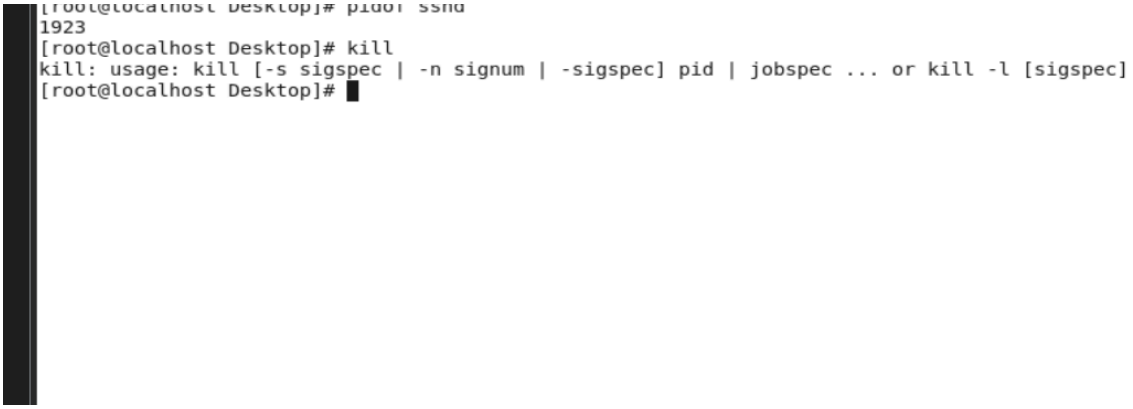
3.pstree

```
        [-u EUIDLIST] [-U UIDLIST] [-G GIDLIST] [-t TERMLIST] [PATTERN]
[root@localhost Desktop]# clear
[root@localhost Desktop]# pstree
init─┬─NetworkManager─┬─dhclient
     │                └─{NetworkManager}
     ├─VGAuthService
     ├─abrtd
     ├─acpid
     ├─anacron
     ├─atd
     ├─auditd───{auditd}
     ├─avahi-daemon───avahi-daemon
     ├─bonobo-activati───{bonobo-activat}
     ├─clock-applet
     ├─console-kit-dae───63*[{console-kit-da}]
     ├─crond
     ├─cupsd
     ├─2*[dbus-daemon───{dbus-daemon}]
     ├─2*[dbus-launch]
     ├─devkit-power-da
     ├─gconfd-2
     ├─gdm-binary───gdm-simple-slav─┬─Xorg
     │                              └─gdm-session-wor───gnome-session─┬─bluetooth-apple
     │                                                               ├─gdu-notificatio
     │                                                               ├─gnome-panel
     │                                                               ├─gnome-power-man
     │                                                               ├─gnome-volume-co
     │                                                               ├─gpk-update-icon
     │                                                               ├─metacity───{metacity}
     │                                                               ├─nautilus
     │                                                               ├─nm-applet
     │                                                               ├─polkit-gnome-au
     │                                                               ├─python
     │                                                               └─{gnome-session}
     ├─gdm-user-switch
     ├─gnome-keyring-d───2*[{gnome-keyring-}]
     ├─gnome-screensav
     ├─gnome-settings-───{gnome-settings}
 root@localhost:~/Des...
```

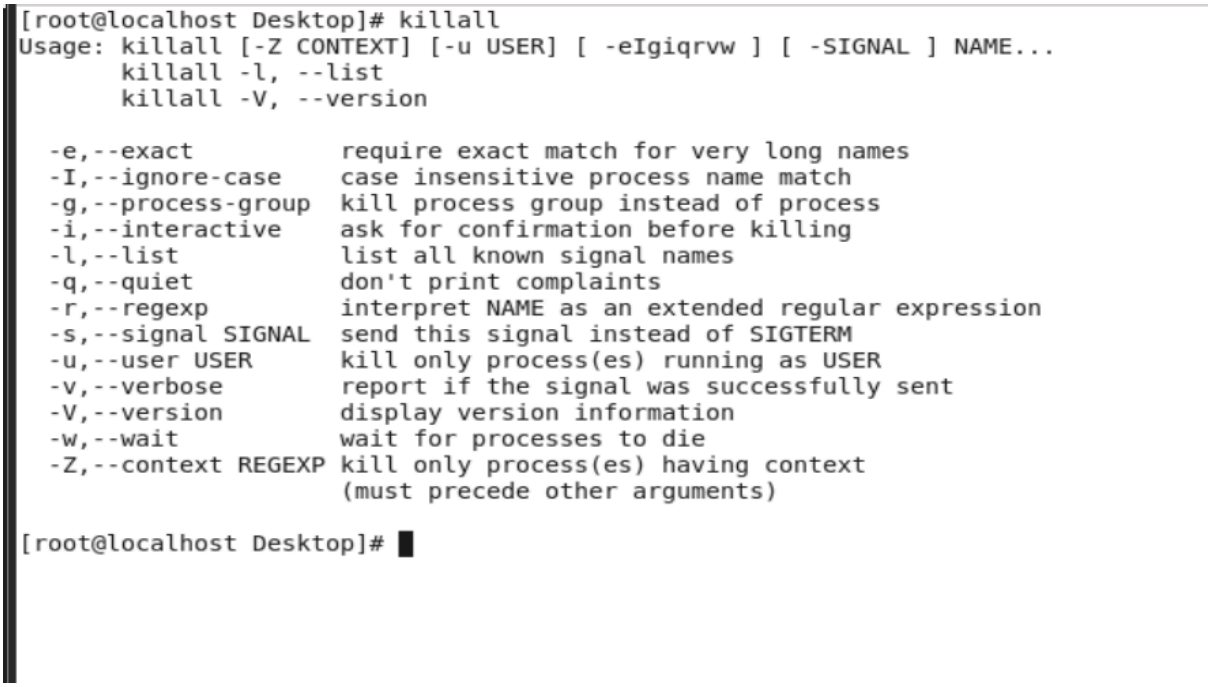
4.Pidof



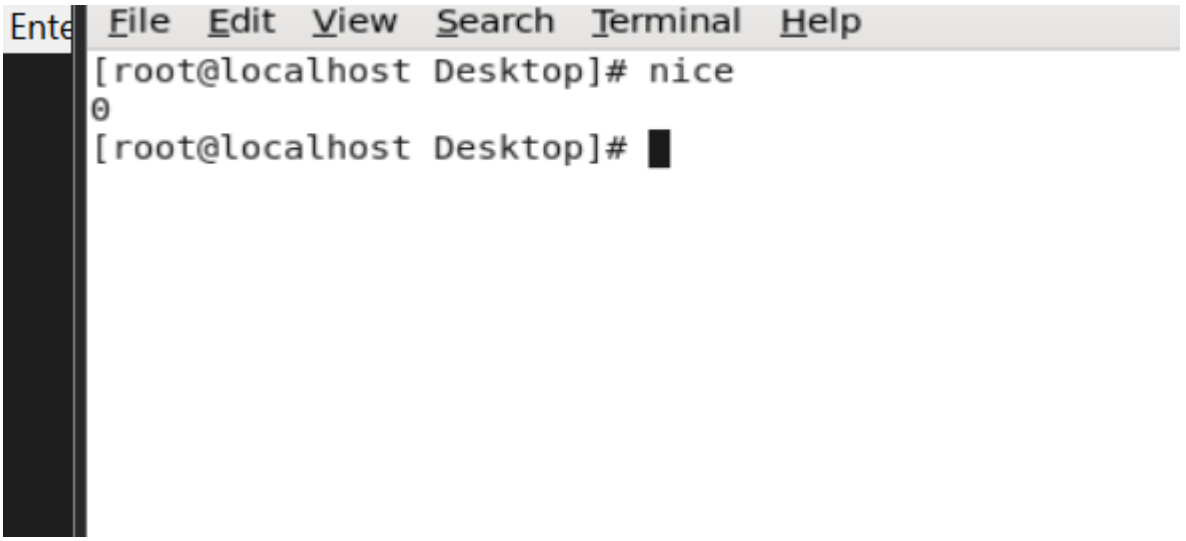
5.kill



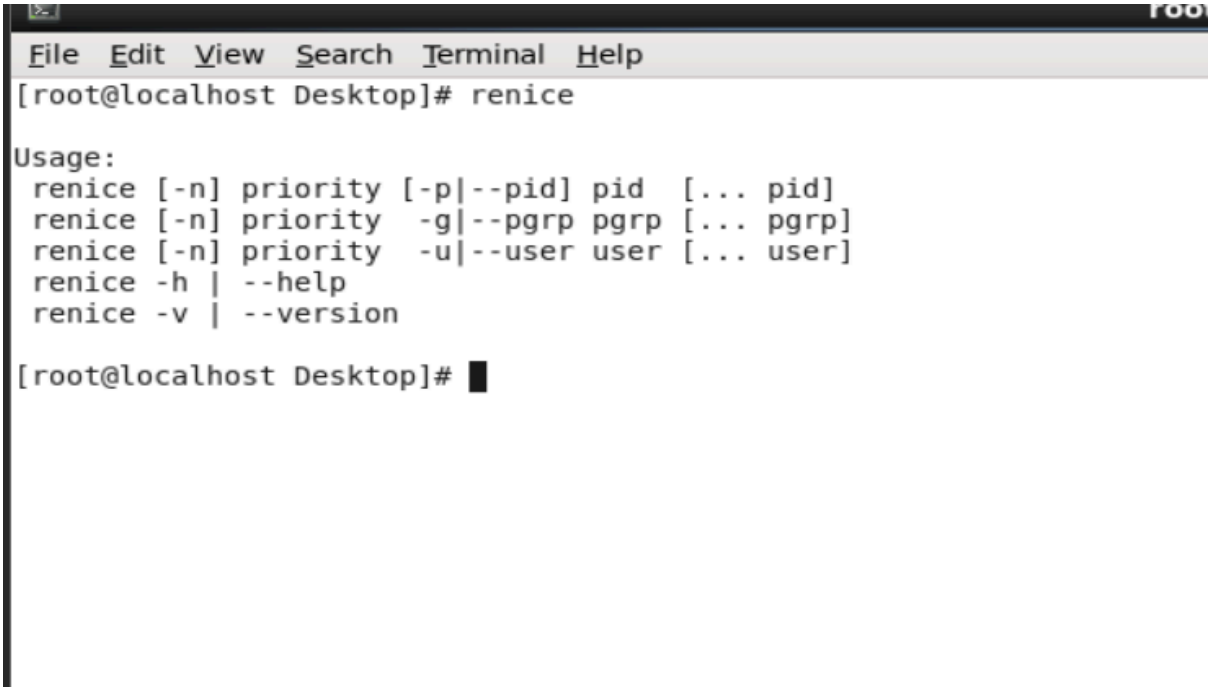
6.killall



7.nice



8.renice



Practical:7. Managing Storage

a. Understanding MBR and GPT Partitions

b. Managing Partitions and File Systems

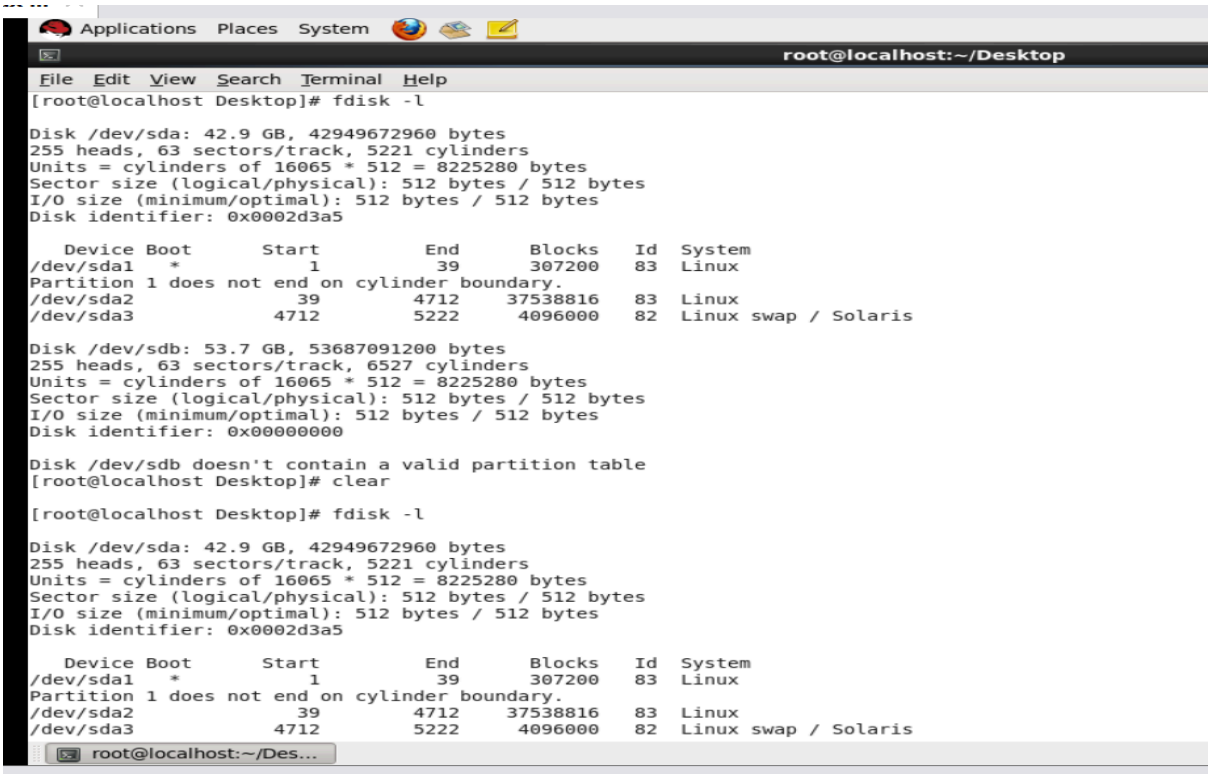
c. Mounting File Systems

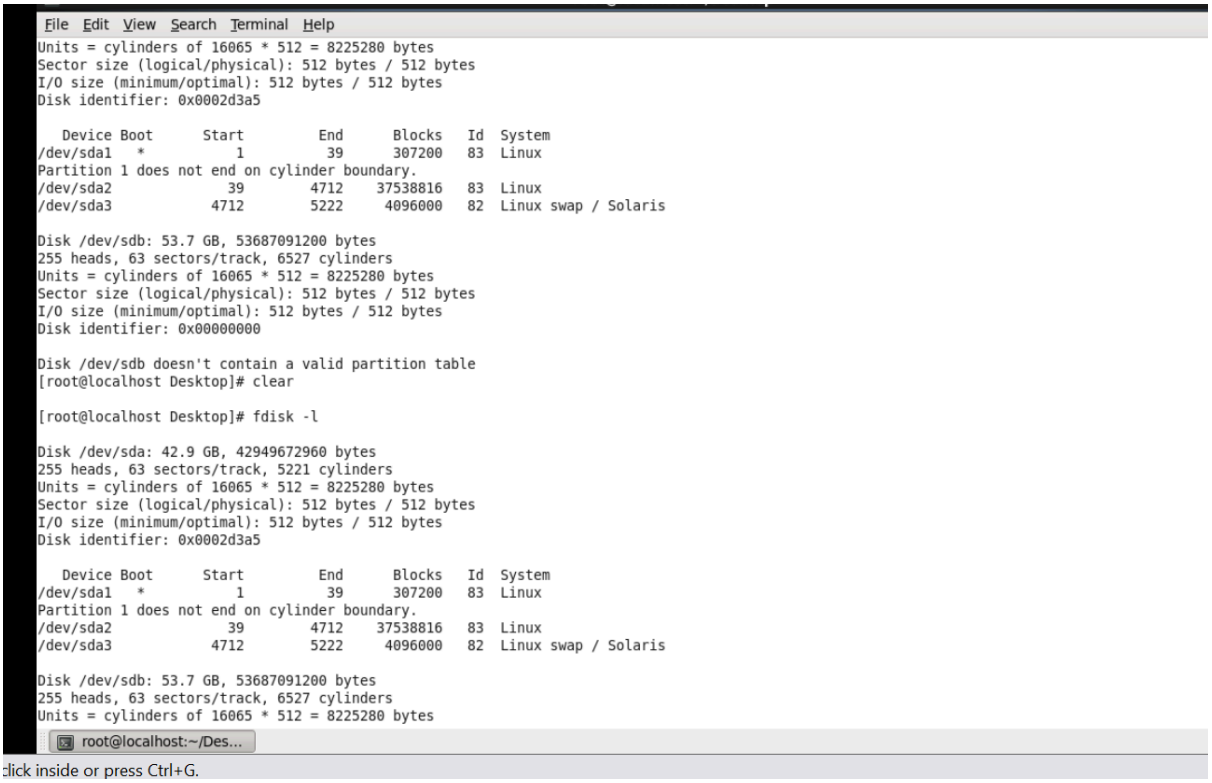
d. List, create, delete partitions on MBR and GPT disks

e. Configure systems to mount file systems at boot by universally unique ID (UUID) or label

f. Add new partitions and logical volumes, and swap to a system non-destructively

g. Create, mount, unmount, and use vfat, ext4, and xfs file systems

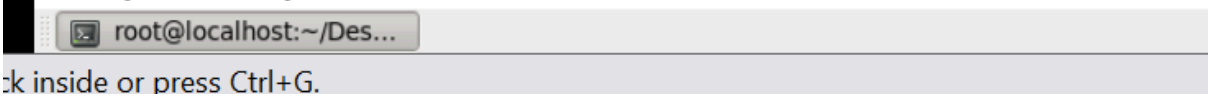




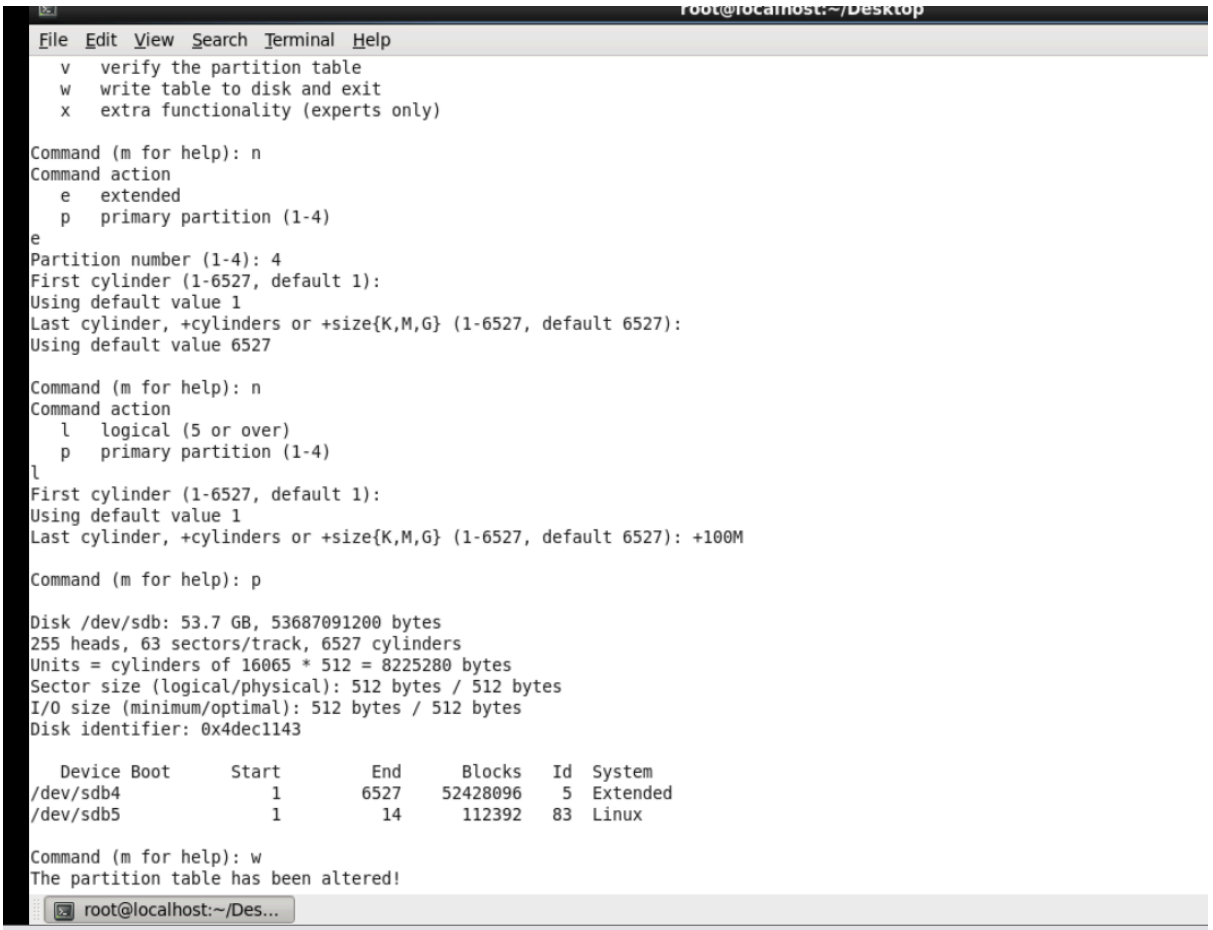
```
Command (m for help): m
Command action
   a   toggle a bootable flag
   b   edit bsd disklabel
   c   toggle the dos compatibility flag
   d   delete a partition
   l   list known partition types
   m   print this menu
   n   add a new partition
   o   create a new empty DOS partition table
   p   print the partition table
   q   quit without saving changes
   s   create a new empty Sun disklabel
   t   change a partition's system id
   u   change display/entry units
   v   verify the partition table
   w   write table to disk and exit
   x   extra functionality (experts only)

Command (m for help): n
Command action
   e   extended
   p   primary partition (1-4)
e
Partition number (1-4): 4
First cylinder (1-6527, default 1):
Using default value 1
Last cylinder, +cylinders or +size{K,M,G} (1-6527, default 6527):
Using default value 6527

Command (m for help): n
Command action
   l   logical (5 or over)
   p   primary partition (1-4)
l
First cylinder (1-6527, default 1):
Using default value 1
Last cylinder, +cylinders or +size{K,M,G} (1-6527, default 6527): +100M
```



ck inside or press Ctrl+G.



ck inside or press Ctrl+G.

Practical:8. Configuring Networking

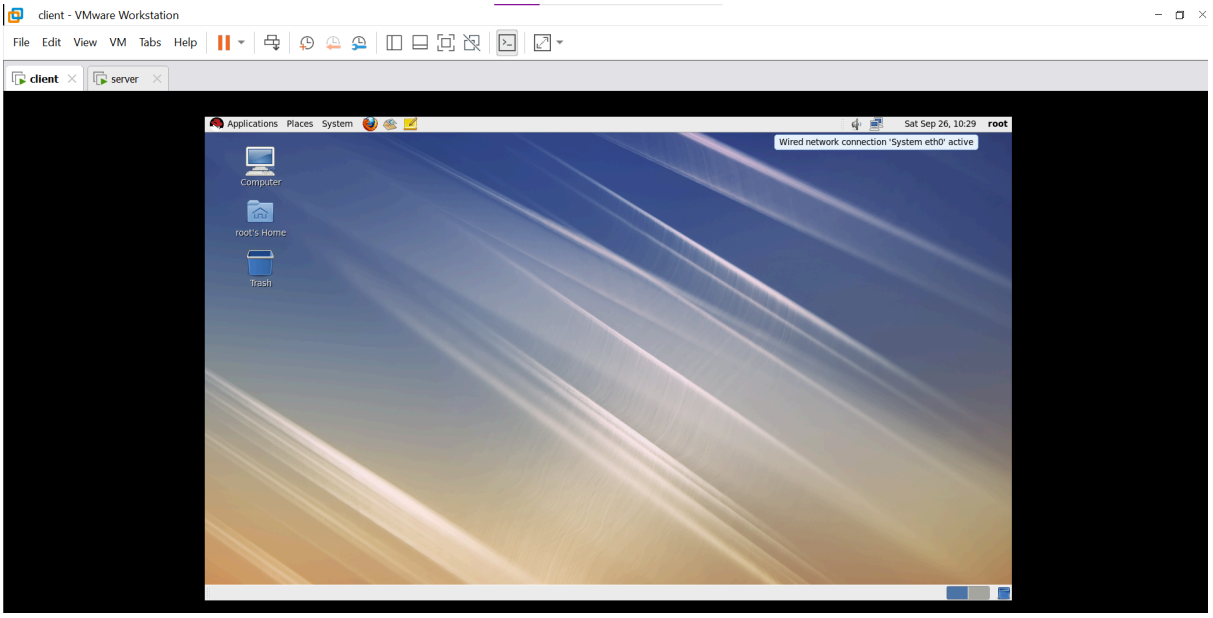
a. Managing Network Addresses and Interfaces

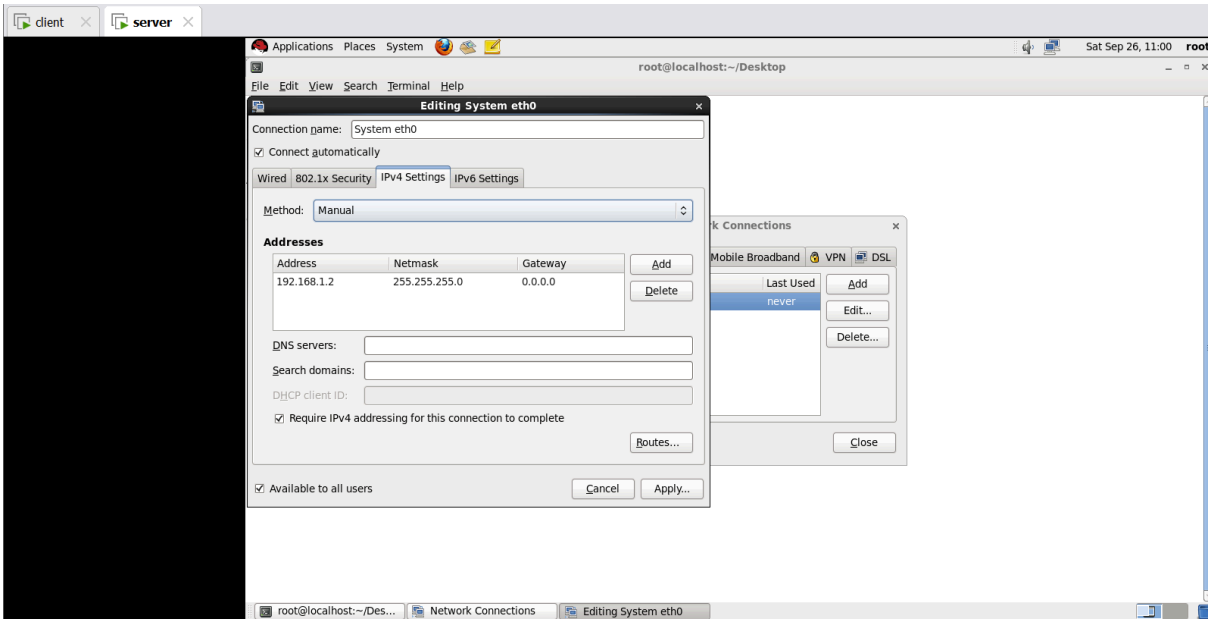
b. Validating Network Configuration

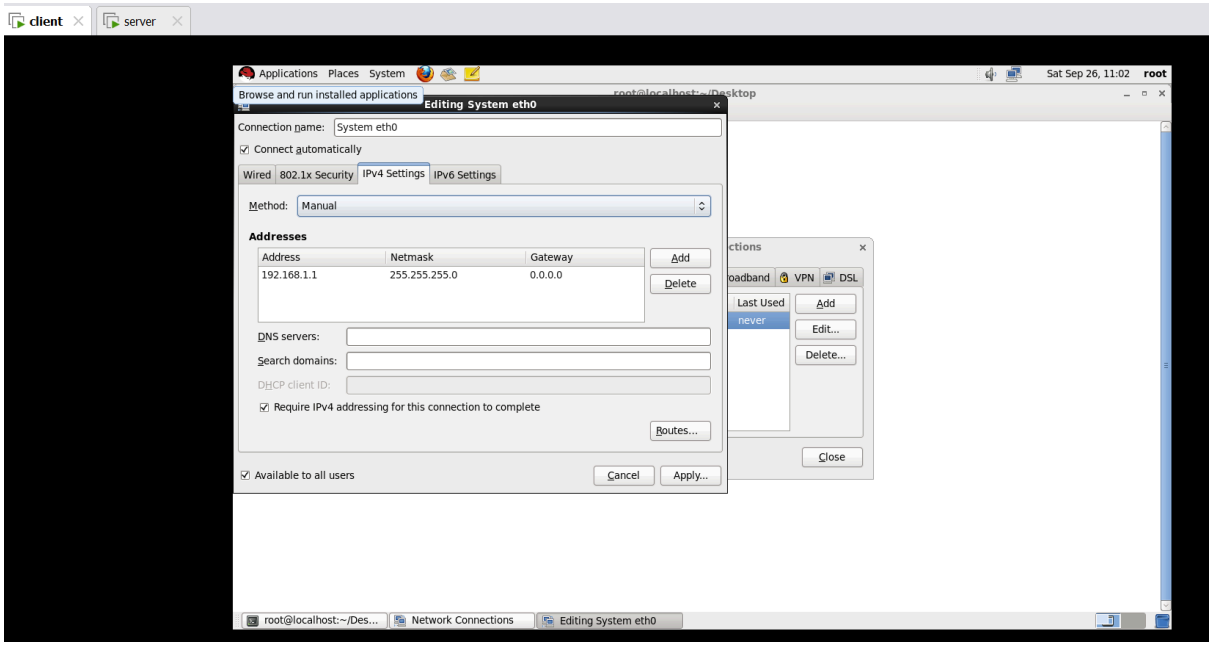
c. Managing Network Configuration with nmtui and nmcli

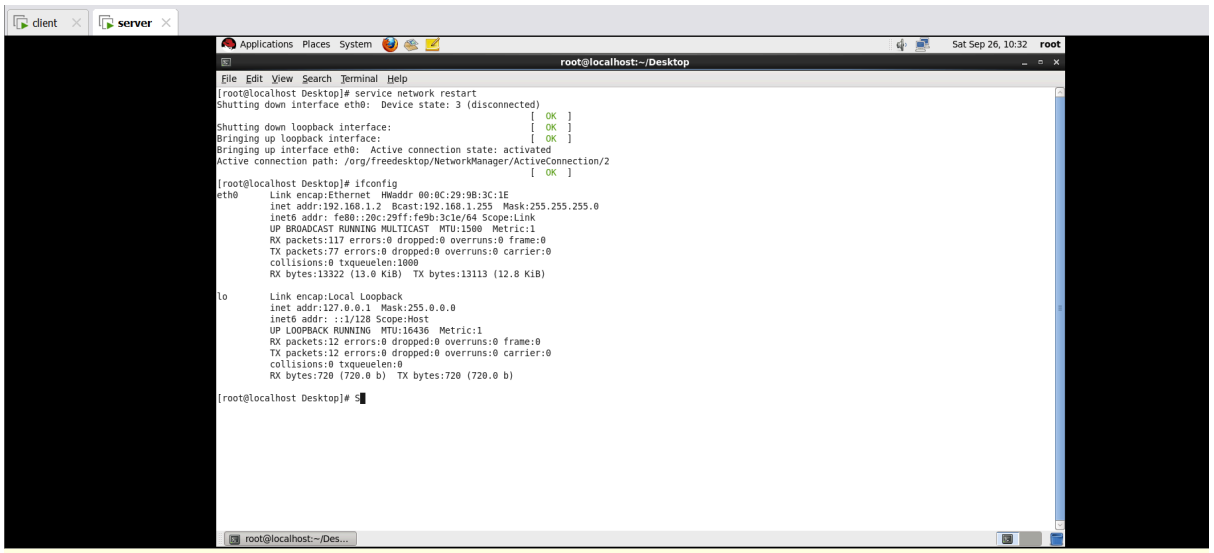
d. Configure IPv4 and IPv6 addresses

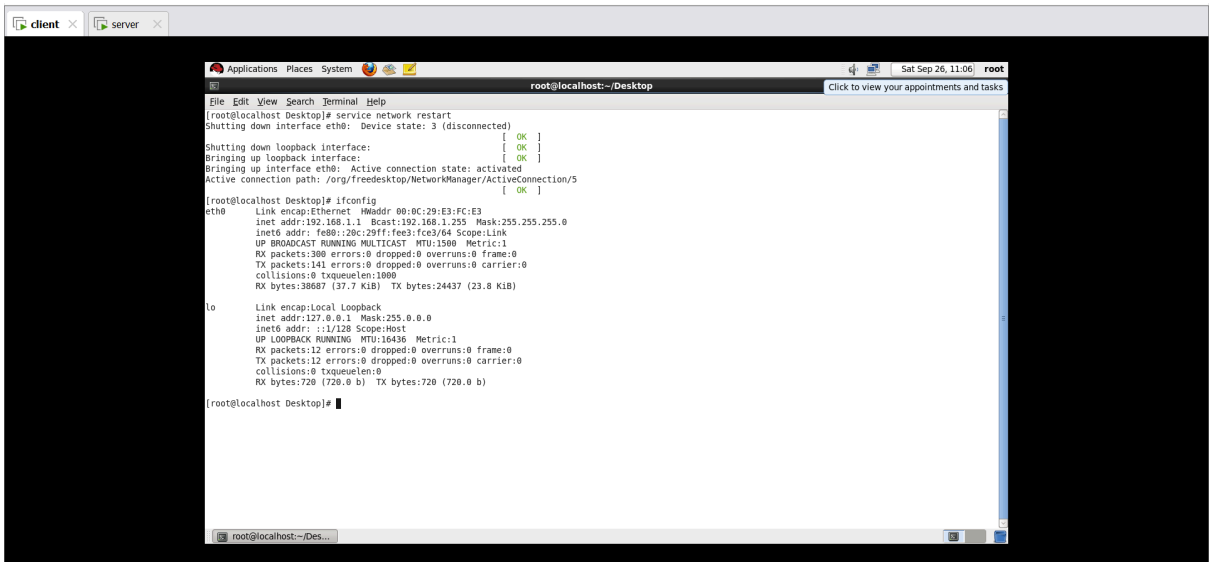
e. Configure DNS-hostname resolution

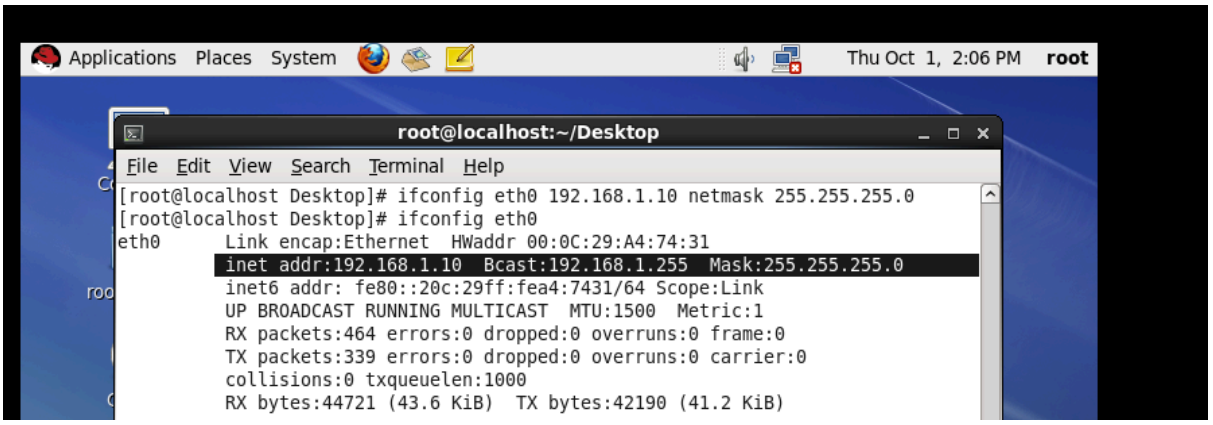


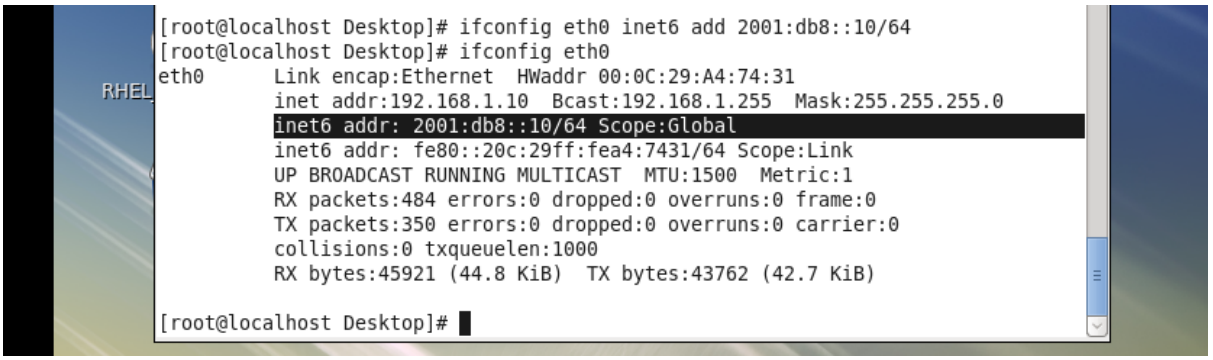



client
server
Applications Places System
Sat Sep 26, 11:02 root
Browse and run installed applications
Editing System eth0
Connection name: System eth0
Connect automatically
Wired 802.1x Security IPv4 Settings IPv6 Settings
Method: Manual
Addresses
Address Netmask Gateway
192.168.1.1 255.255.255.0 0.0.0.0
Add
Delete
DNS servers:
Search domains:
DHCP client ID:
Require IPv4 addressing for this connection to complete
Routes...
Available to all users
Cancel
Apply...
Broadband VPN DSL
Last Used
never
Add
Edit...
Delete...
Close
root@localhost:~/Des...
Network Connections
Editing System eth0


client
server
Applications Places System
Sat Sep 26, 10:32 root
root@localhost:~/Desktop
File Edit View Search Terminal Help
[root@localhost Desktop]# service network restart
Shutting down interface eth0: Device state: 3 (disconnected)
[OK]
Shutting down loopback interface: [OK]
Bringing up loopback interface: [OK]
Bringing up interface eth0: Active connection state: activated
Active connection path: /org/freedesktop/NetworkManager/ActiveConnection/2
[OK]
[root@localhost Desktop]# ifconfig
eth0 Link encap:Ethernet HWaddr 00:0C:29:9B:3C:1E
inet addr:192.168.1.2 Bcast:192.168.1.255 Mask:255.255.255.0
inet6 addr: fe80::20c:29ff:fe9b:3c1e/64 Scope:Link
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:117 errors:0 dropped:0 overruns:0 frame:0
TX packets:77 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:13322 (13.0 KiB) TX bytes:13113 (12.8 KiB)
lo Link encap:Local Loopback
inet addr:127.0.0.1 Mask:255.0.0.0
inet6 addr: ::1/128 Scope:Host
UP LOOPBACK RUNNING MTU:16436 Metric:1
RX packets:12 errors:0 dropped:0 overruns:0 frame:0
TX packets:12 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:720 (720.0 b) TX bytes:720 (720.0 b)
[root@localhost Desktop]# s
root@localhost:~/Des...


client
server
Applications Places System
Sat Sep 26, 11:06 root
root@localhost:~/Desktop
Click to view your appointments and tasks
File Edit View Search Terminal Help
[root@localhost Desktop]# service network restart
Shutting down interface eth0: Device state: 3 (disconnected)
[OK]
Shutting down loopback interface: [OK]
Bringing up loopback interface: [OK]
Bringing up interface eth0: Active connection state: activated
Active connection path: /org/freedesktop/NetworkManager/ActiveConnection/5
[OK]
[root@localhost Desktop]# ifconfig
eth0 Link encap:Ethernet HWaddr 00:0C:29:E3:FC:E3
inet addr:192.168.1.1 Bcast:192.168.1.255 Mask:255.255.255.0
inet6 addr: fe80::20c:29ff:fee3:fce3/64 Scope:Link
UP BROADCAST RUNNING MULTICAST MTU:1500 Metric:1
RX packets:300 errors:0 dropped:0 overruns:0 frame:0
TX packets:141 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:1000
RX bytes:38687 (37.7 KiB) TX bytes:24437 (23.8 KiB)
lo Link encap:Local Loopback
inet addr:127.0.0.1 Mask:255.0.0.0
inet6 addr: ::1/128 Scope:Host
UP LOOPBACK RUNNING MTU:16436 Metric:1
RX packets:12 errors:0 dropped:0 overruns:0 frame:0
TX packets:12 errors:0 dropped:0 overruns:0 carrier:0
collisions:0 txqueuelen:0
RX bytes:720 (720.0 b) TX bytes:720 (720.0 b)
[root@localhost Desktop]#
root@localhost:~/Des...

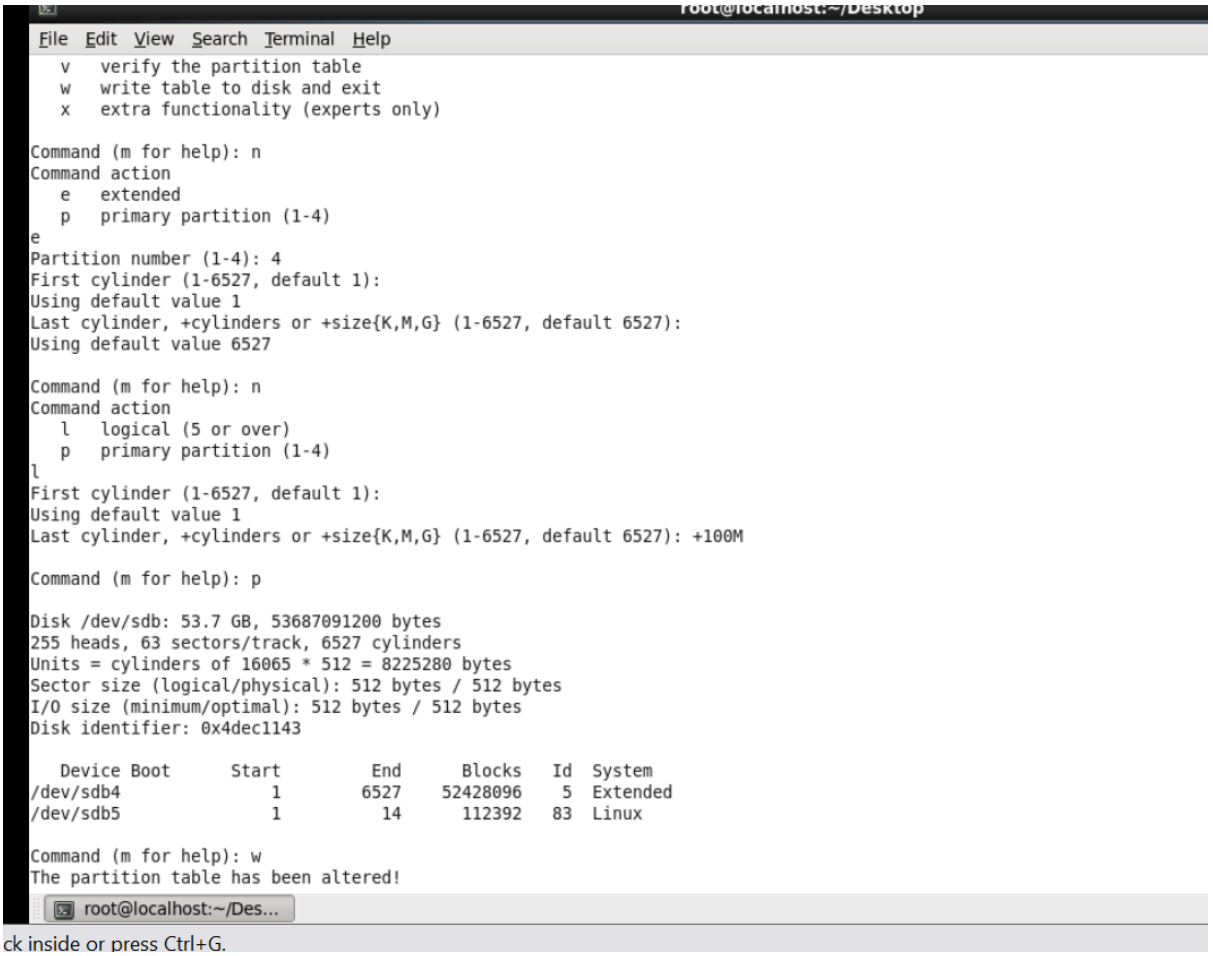





```
Command (m for help): m
Command action
   a   toggle a bootable flag
   b   edit bsd disklabel
   c   toggle the dos compatibility flag
   d   delete a partition
   l   list known partition types
   m   print this menu
   n   add a new partition
   o   create a new empty DOS partition table
   p   print the partition table
   q   quit without saving changes
   s   create a new empty Sun disklabel
   t   change a partition's system id
   u   change display/entry units
   v   verify the partition table
   w   write table to disk and exit
   x   extra functionality (experts only)

Command (m for help): n
Command action
   e   extended
   p   primary partition (1-4)
e
Partition number (1-4): 4
First cylinder (1-6527, default 1):
Using default value 1
Last cylinder, +cylinders or +size{K,M,G} (1-6527, default 6527):
Using default value 6527

Command (m for help): n
Command action
   l   logical (5 or over)
   p   primary partition (1-4)
l
First cylinder (1-6527, default 1):
Using default value 1
Last cylinder, +cylinders or +size{K,M,G} (1-6527, default 6527): +100M
```



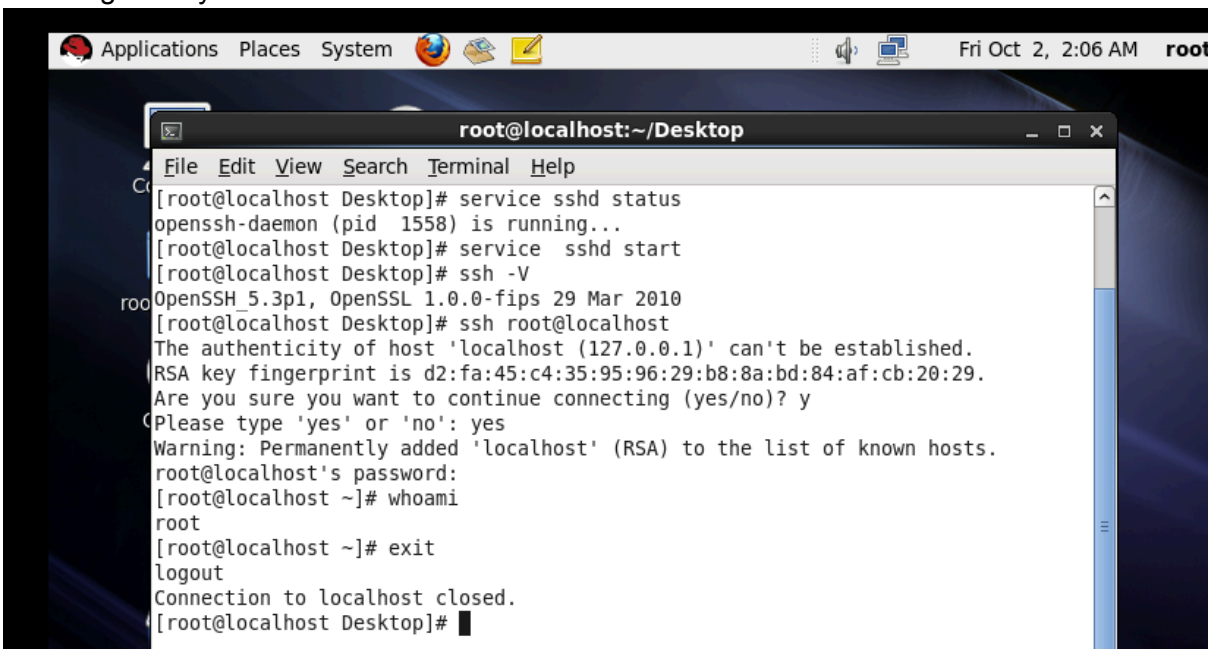
9. Configuring SSH

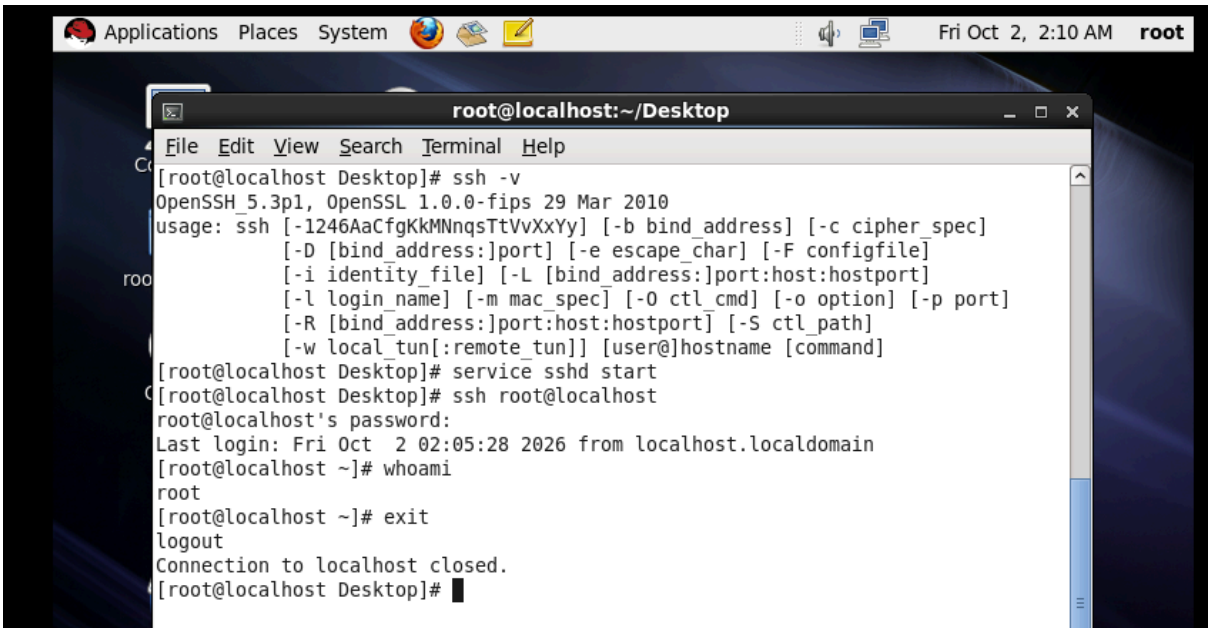
a. Using SSH and Related Utilities

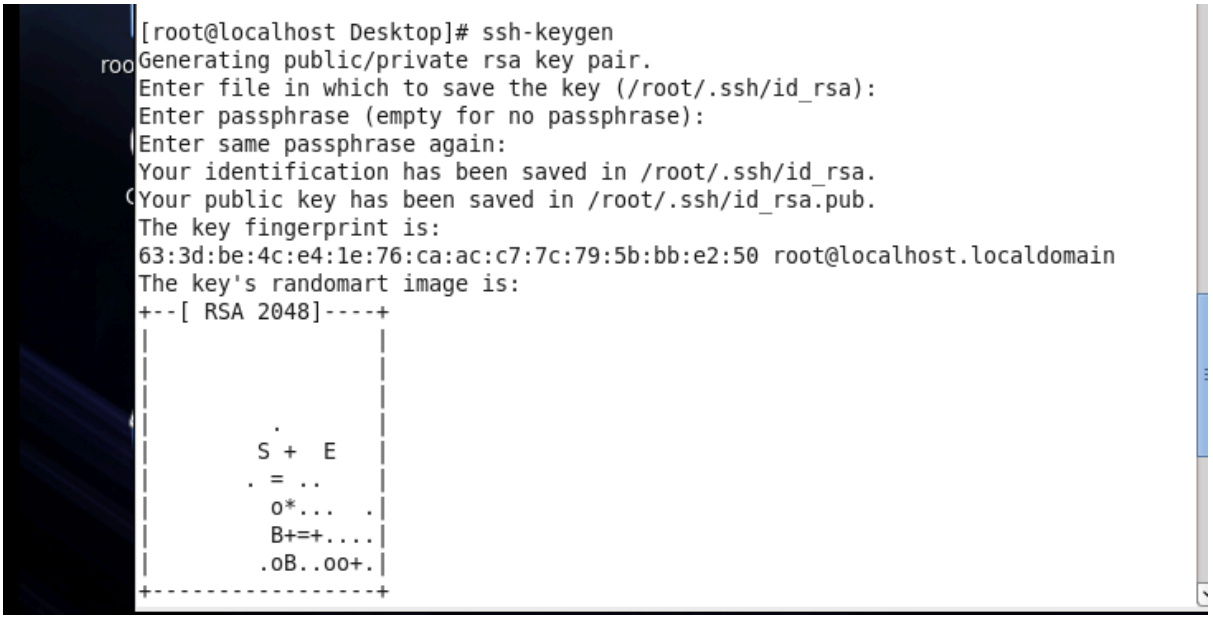
b. Access remote systems using SSH

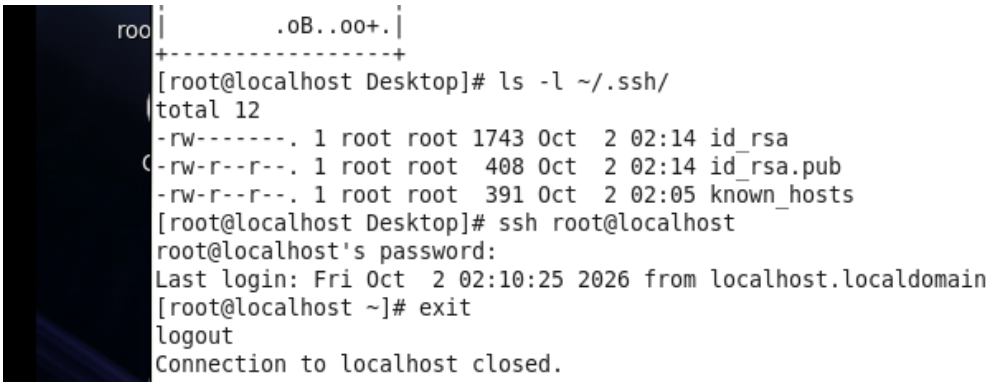
c. Configuring Key-Based Authentication with Passphrases

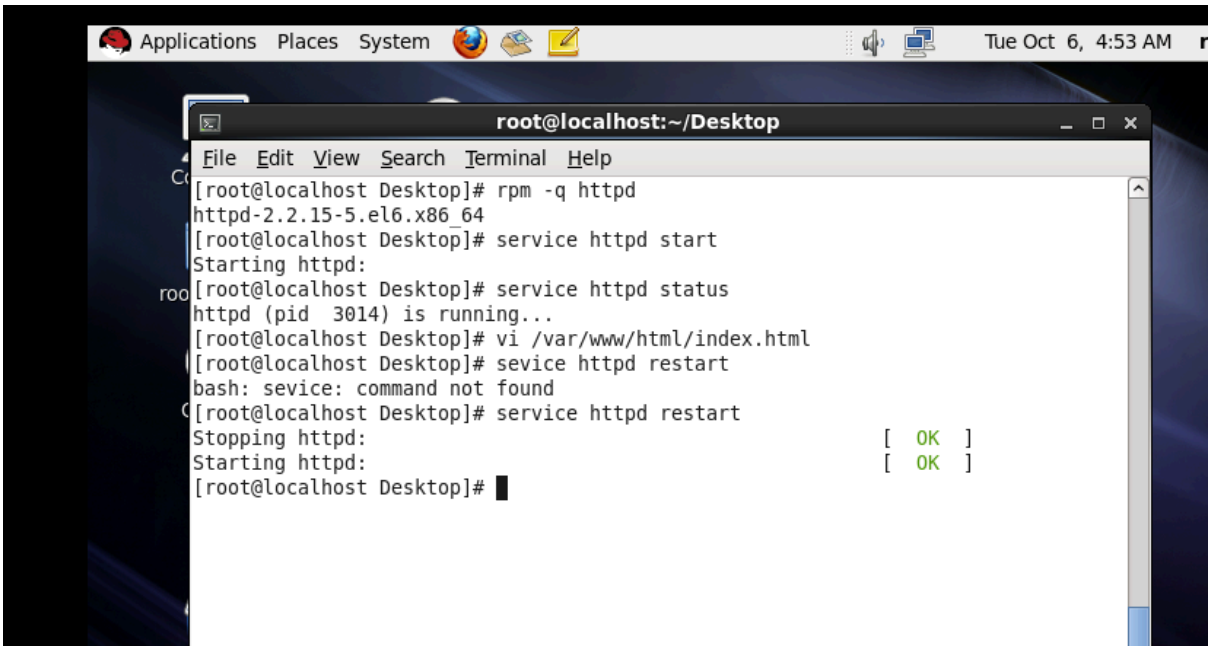
d. Configure key-based authentication for SSH

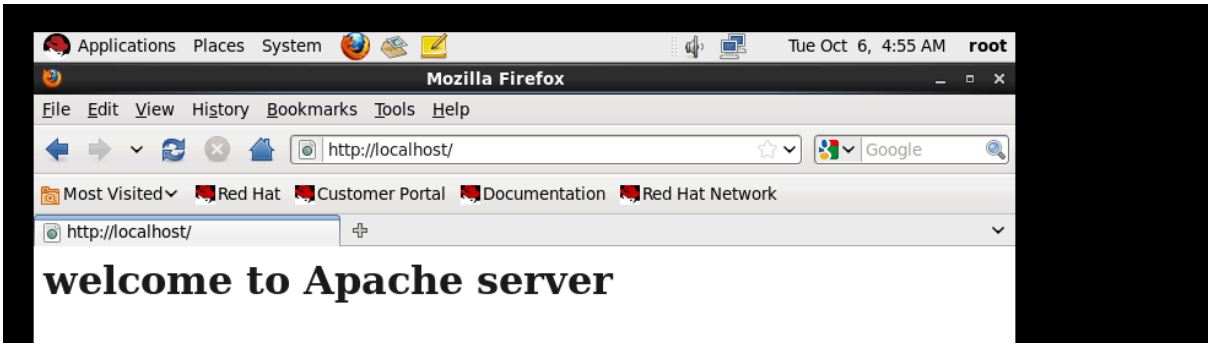


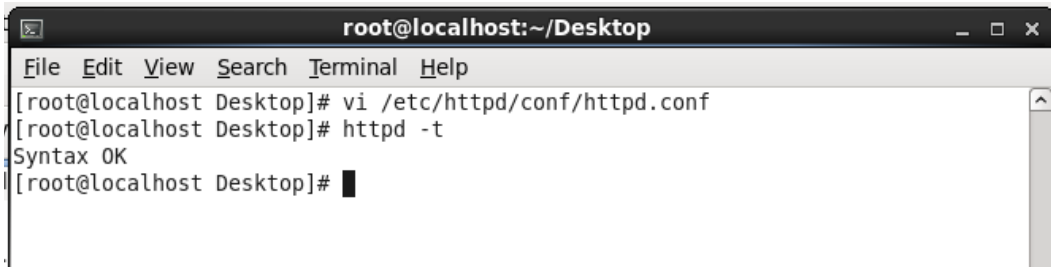

Applications Places System
Fri Oct 2, 2:10 AM root
root@localhost:~/Desktop
File Edit View Search Terminal Help
[root@localhost Desktop]# ssh -v
OpenSSH_5.3p1, OpenSSL 1.0.0-fips 29 Mar 2010
usage: ssh [-1246AaCfgKkMNnqsTtVvXxYy] [-b bind_address] [-c cipher_spec]
[-D [bind_address:]port] [-e escape_char] [-F configfile]
[-i identity_file] [-L [bind_address:]port:host:hostport]
[-l login_name] [-m mac_spec] [-O ctl_cmd] [-o option] [-p port]
[-R [bind_address:]port:host:hostport] [-S ctl_path]
[-w local_tun[:remote_tun]] [user@]hostname [command]
[root@localhost Desktop]# service sshd start
[root@localhost Desktop]# ssh root@localhost
root@localhost's password:
Last login: Fri Oct 2 02:05:28 2026 from localhost.localdomain
[root@localhost ~]# whoami
root
[root@localhost ~]# exit
logout
Connection to localhost closed.
[root@localhost Desktop]#

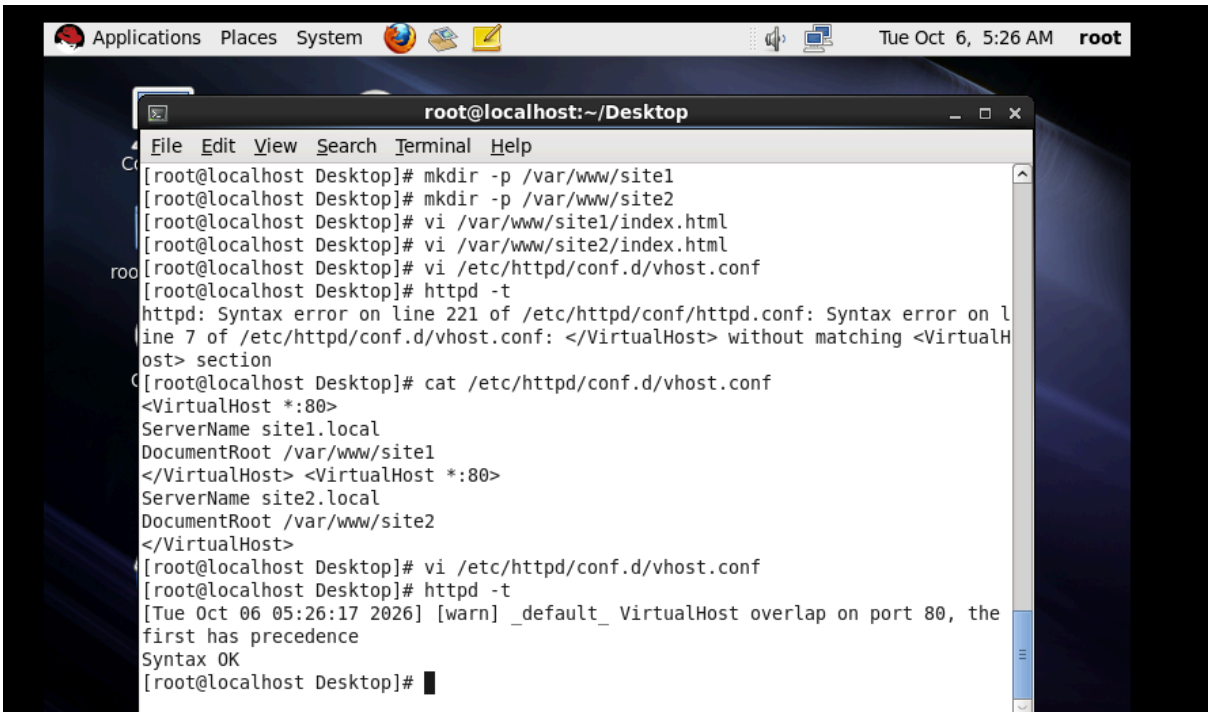

[root@localhost Desktop]# ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/root/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /root/.ssh/id_rsa.
Your public key has been saved in /root/.ssh/id_rsa.pub.
The key fingerprint is:
63:3d:be:4c:e4:1e:76:ca:ac:c7:7c:79:5b:bb:e2:50 root@localhost.localdomain
The key's randomart image is:
+--[RSA 2048]----+
| . |
| S + E |
| . = .. |
| o*... . |
| B+=+.... |
| .oB..oo+. |
+-----------------+

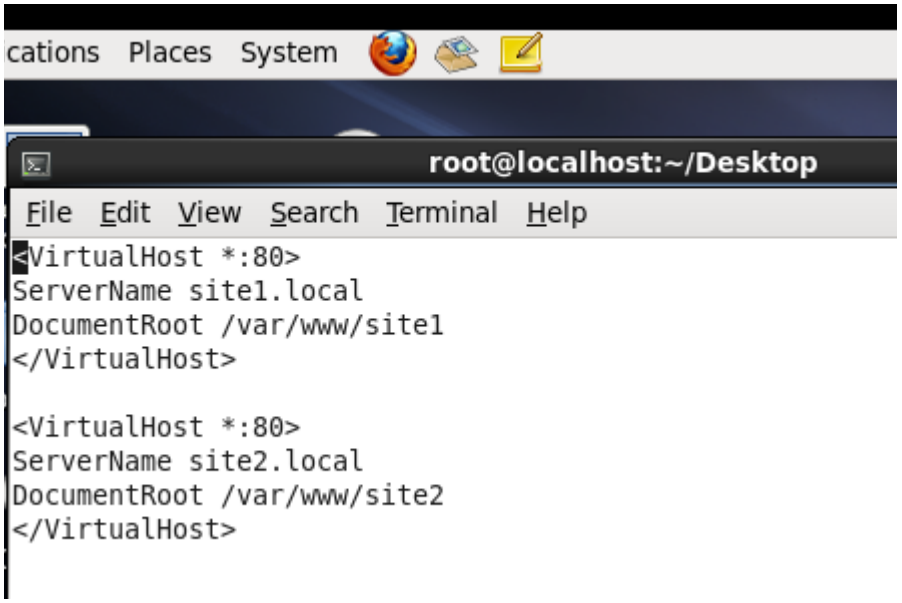

| .oB..oo+. |
+-----------------+
[root@localhost Desktop]# ls -l ~/.ssh/
total 12
-rw-------. 1 root root 1743 Oct 2 02:14 id_rsa
-rw-r--r--. 1 root root 408 Oct 2 02:14 id_rsa.pub
-rw-r--r--. 1 root root 391 Oct 2 02:05 known_hosts
[root@localhost Desktop]# ssh root@localhost
root@localhost's password:
Last login: Fri Oct 2 02:10:25 2026 from localhost.localdomain
[root@localhost ~]# exit
logout
Connection to localhost closed.

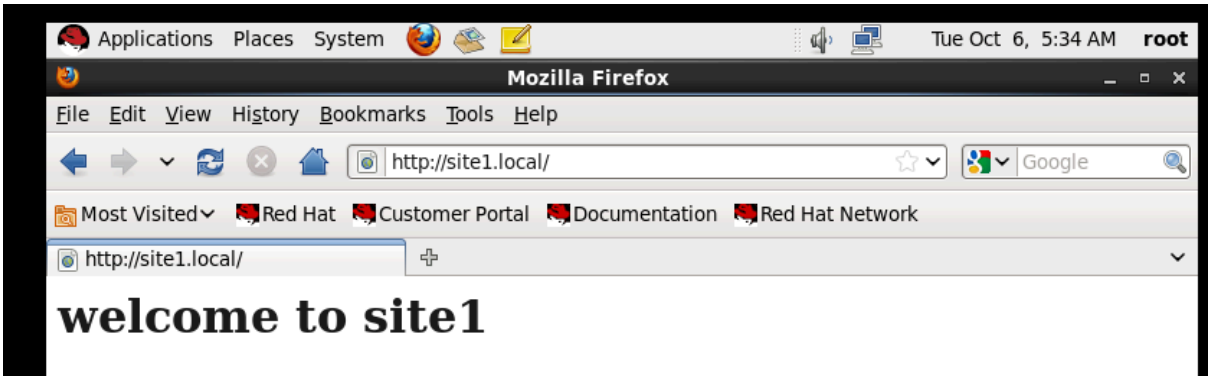

Applications Places System
Tue Oct 6, 4:53 AM
root@localhost:~/Desktop
File Edit View Search Terminal Help
[root@localhost Desktop]# rpm -q httpd
httpd-2.2.15-5.el6.x86_64
[root@localhost Desktop]# service httpd start
Starting httpd:
[root@localhost Desktop]# service httpd status
httpd (pid 3014) is running...
[root@localhost Desktop]# vi /var/www/html/index.html
[root@localhost Desktop]# sevice httpd restart
bash: sevice: command not found
[root@localhost Desktop]# service httpd restart
Stopping httpd: [OK]
Starting httpd: [OK]
[root@localhost Desktop]#


Applications Places System
Tue Oct 6, 4:55 AM root
Mozilla Firefox
File Edit View History Bookmarks Tools Help
http://localhost/
Google
Most Visited Red Hat Customer Portal Documentation Red Hat Network
http://localhost/
welcome to Apache server


root@localhost:~/Desktop
File Edit View Search Terminal Help
[root@localhost Desktop]# vi /etc/httpd/conf/httpd.conf
[root@localhost Desktop]# httpd -t
Syntax OK
[root@localhost Desktop]#


Applications Places System
Tue Oct 6, 5:26 AM root
root@localhost:~/Desktop
File Edit View Search Terminal Help
[root@localhost Desktop]# mkdir -p /var/www/site1
[root@localhost Desktop]# mkdir -p /var/www/site2
[root@localhost Desktop]# vi /var/www/site1/index.html
[root@localhost Desktop]# vi /var/www/site2/index.html
[root@localhost Desktop]# vi /etc/httpd/conf.d/vhost.conf
[root@localhost Desktop]# httpd -t
httpd: Syntax error on line 221 of /etc/httpd/conf/httpd.conf: Syntax error on l
ine 7 of /etc/httpd/conf.d/vhost.conf: </VirtualHost> without matching <VirtualH
ost> section
[root@localhost Desktop]# cat /etc/httpd/conf.d/vhost.conf
<VirtualHost *:80>
ServerName site1.local
DocumentRoot /var/www/site1
</VirtualHost> <VirtualHost *:80>
ServerName site2.local
DocumentRoot /var/www/site2
</VirtualHost>
[root@localhost Desktop]# vi /etc/httpd/conf.d/vhost.conf
[root@localhost Desktop]# httpd -t
[Tue Oct 06 05:26:17 2026] [warn] _default_ VirtualHost overlap on port 80, the
first has precedence
Syntax OK
[root@localhost Desktop]#


cations Places System
root@localhost:~/Desktop
File Edit View Search Terminal Help
<VirtualHost *:80>
ServerName site1.local
DocumentRoot /var/www/site1
</VirtualHost>
<VirtualHost *:80>
ServerName site2.local
DocumentRoot /var/www/site2
</VirtualHost>


Applications Places System
Tue Oct 6, 5:34 AM root
Mozilla Firefox
File Edit View History Bookmarks Tools Help
http://site1.local/
Google
Most Visited Red Hat Customer Portal Documentation Red Hat Network
http://site1.local/
welcome to site1

```
[root@localhost Desktop]# service iptables status
Table: filter
Chain INPUT (policy ACCEPT)
num  target     prot opt source               destination
1    ACCEPT     all  --  0.0.0.0/0            0.0.0.0/0           state RELATED,
ESTABLISHED
2    ACCEPT     icmp --  0.0.0.0/0            0.0.0.0/0
3    ACCEPT     all  --  0.0.0.0/0            0.0.0.0/0
4    ACCEPT     tcp  --  0.0.0.0/0            0.0.0.0/0           state NEW tcp
dpt:22
5    REJECT     all  --  0.0.0.0/0            0.0.0.0/0           reject-with ic
mp-host-prohibited

Chain FORWARD (policy ACCEPT)
num  target     prot opt source               destination
1    REJECT     all  --  0.0.0.0/0            0.0.0.0/0           reject-with ic
mp-host-prohibited

Chain OUTPUT (policy ACCEPT)
num  target     prot opt source               destination
```

```
[root@localhost Desktop]# iptables -L
Chain INPUT (policy ACCEPT)
target     prot opt source               destination
ACCEPT     all  --  anywhere             anywhere             state RELATE
LISHED
```

```
hain OUTPUT (policy ACCEPT)
arget     prot opt source               destination
root@localhost Desktop]# iptable -A INPUT -p tcp --dport 22 -j ACCEPT
ash: iptable: command not found
root@localhost Desktop]# iptables -A INPUT -p tcp --dport 22 -j ACCEPT
root@localhost Desktop]# iptables -L -n
hain INPUT (policy ACCEPT)
arget     prot opt source               destination
CCEPT     all  --  0.0.0.0/0            0.0.0.0/0           state RELATED,ESTAB
ISHED
CCEPT     icmp --  0.0.0.0/0            0.0.0.0/0
CCEPT     all  --  0.0.0.0/0            0.0.0.0/0
CCEPT     tcp  --  0.0.0.0/0            0.0.0.0/0           state NEW tcp dpt:2

EJECT     all  --  0.0.0.0/0            0.0.0.0/0           reject-with icmp-ho
t-prohibited
```